

Review

Degradation of microplastics in the natural environment: A comprehensive review on process, mechanism, influencing factor and leaching behavior

Hongwei Luo^{a,b,*}, Zhen Li^a, Yifeng Zeng^a, Dongqin He^a, Jianqiang Sun^a, Anping Zhang^a, Xiangliang Pan^{a,**}

^a College of Environment, Zhejiang University of Technology, Hangzhou, 310014, China

^b Shaoxing Research Institute, Zhejiang University of Technology, Shaoxing, 312085, China

ARTICLE INFO

Keywords:

Microplastics

Degradation

Additives

Influencing factors

AOPs

ABSTRACT

In recent years, the pollution of microplastics (MPs) has attracted global attention because of their extensive distribution and significant impact on biota. Once MPs enter the ecosystem, the complex degradation process that MPs undergo will alter their surface properties and toxic effects. In order to address the global pollution issue caused by MPs, a range of technologies have been developed to degrade MPs. This study systematically summarizes the degradation methods of MPs reported recently, including physical, chemical and biological degradation processes. Dissolved organic matter (DOM), some reducing agents and sulfides are found to cause MPs aging and degradation. The roles of reactive oxygen species and biological enzymes in MPs degradation are elucidated. The chemical and biological degradation of various types of MPs are discussed and their degradation pathways are clarified. A number of factors affect the rates of MPs degradation in natural environment. Specifically, the influences of external factors and the intrinsic properties of MPs on their degradation are revealed. Furthermore, the potential environmental toxicity of MPs is investigated in greater depth, with particular attention paid to the release of plastic additives and DOM during the degradation process. This review article will assist in the comprehension of environmental degradation process and ecological risks of MPs and provide guidance for controlling the environmental pollution caused by MPs.

1. Introduction

Plastic products are advantageous in numerous daily activities due to their light weight, excellent chemical stability and low production costs (Guo and Wang, 2019). Statistical data showed that the annual production of plastic products in 2018 was 360 million tons, with an anticipated increase to 500 million tons by 2025 (Bai et al., 2019). Traditional plastics are considered persistent pollutants due to their ability to remain in aquatic environments for extended periods. These plastics are often considered non-biodegradable. Plastic products may break down into tiny plastic particles in natural environment due to a combination of physical, chemical and biological processes. Plastics are widely distributed in environmental media, and plastic particles less than 5 mm in diameter are called microplastics (MPs) (Harris, 2020). MPs have been detected in nearly every environmental compartment, from remote mountain air (Allen et al., 2022) to deep ocean sediments (Peng et al., 2018). Recent studies show that surface waters of the

Yangtze River have MPs concentrations ranging from 1000 to 10,000 particles/m³ (Zhang et al., 2016). MPs have a large number and small size, so they can easily enter human body through a variety of ways. As a result, MPs are more harmful to humans compared to larger plastics (García-Montelongo et al., 2014). Many studies worldwide show that MPs have serious effects on organism growth, reproduction, and mortality (Elizalde-Velázquez and Gómez-Oliván, 2021). So far, the pollution of MPs has emerged as a global environmental concern.

MPs will undergo severe degradation processes in the natural environment, including physical, chemical and biological degradation. Researchers have identified significant alterations in the physical and chemical characteristics of degraded MPs, which affected their interaction with the environment. For example, an increased specific surface area offers greater opportunities for contaminants to adhere (Liu et al., 2019a). Researchers found that naturally degraded MPs from the ocean not only produce oxygen-containing groups, but also change their hydrophilicity and surface charge, thereby affecting the interaction

* Corresponding author. College of Environment, Zhejiang University of Technology, Hangzhou, 310014, China.

** Corresponding author.

E-mail addresses: hwluo@zjut.edu.cn (H. Luo), panxl@zjut.edu.cn (X. Pan).

<https://doi.org/10.1016/j.jenvman.2025.126429>

Received 7 March 2025; Received in revised form 12 June 2025; Accepted 27 June 2025

Available online 1 July 2025

0301-4797/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

between MPs and coexisting contaminants (Andrady, 2022). Many researchers conduct laboratory simulations to explore how the physicochemical and toxicological characteristics of MPs change under varying environmental conditions, including physical wear, chemical degradation and biological degradation. At present, there are a large number of reviews focusing on the characterization techniques and exploring degradation processes of MPs under various conditions (Alimi et al., 2018; Andrady, 2017; Enfrin et al., 2021). However, there is currently a lack of review on the leaching behavior of additives and dissolved organic matter (DOM) during MPs degradation. In order to address this gap, understanding the degradation process and potential impacts of MPs is crucial. This review mainly focuses on the natural degradation processes of MPs in aquatic environments, including photodegradation, biodegradation and physical weathering. Artificial degradation techniques (e.g., electrochemistry and advanced oxidation processes (AOPs)) are mentioned briefly for comparison. The environmental fate of MPs under natural conditions remains a central theme.

MPs degradation in natural environment is very complicated because many environmental factors can affect the degradation process. For instance, researchers have demonstrated that environmental components can interfere with photochemical reactions that affect MPs degradation. Therefore, the photodegradation process of MPs is different in different water conditions (Ge et al., 2023; Liu et al., 2020). The selection of a catalyst, the conditions of exposure (pH, temperature, oxygen, light, bioturbation, etc.) and the use of environmental additives have all been identified as playing an important role in this process. These factors often do not act alone, but interact with each other. For example, some researchers have observed that low temperature and pH are more conducive to the photocatalytic degradation of polyethylene (PE) (Ariza-Tarazona et al., 2020). However, others have pointed out that low temperature can counteract the inhibitory effect of low pH on the biodegradation of PE (Dümichen et al., 2017). In addition to the external factors mentioned above, the existence form, structure and additives of plastics can also affect the degradation process of MPs to varying degrees. These findings indicate that complex interactions between degradation factors, and their specific mechanisms need to be further explored.

During the manufacturing and molding process of plastics, various additives are blended into the polymer to obtain different functions of plastics (Luo et al., 2022a). These additives commonly include antioxidants, colorants, filler, antibacterial, plasticizers and stabilizer. Several studies have shown that certain additives can be released from plastics into the environment (Hahladakis et al., 2018; Hermabessiere et al., 2017). Additionally, additives may be degraded and form other toxic compounds after leaching (Costa et al., 2023). Researchers found a fivefold increase in the amount of phthalate esters (PAEs) released from poly (vinyl chloride) (PVC) cables when they were exposed to both light and bacteria compared with either factor alone (Paluselli et al., 2018). MPs with a diameter of less than 130 μm may be capable of traversing human tissues, triggering local immune responses and releasing organic matter. These organic compounds include additives and adsorbed pollutants from the environment (Yu et al., 2019). Previous studies have shown that ingesting plastics not only causes physical and mechanical damage to fish, but also damages their digestive tract and enzyme production. This will further induce oxidative stress in fish, reduce their growth rate, and even affect their reproduction (Jin et al., 2018). In addition, MPs can persist in fish gut and interact with environmental microorganisms (Grigorakis et al., 2017). So far, the leaching mechanism of organic additives from degraded MPs has not been found. The interaction between leached additives and the environment has not been fully documented.

This work systematically reviews recent advances in environmental degradation of MPs in recent years. The objective is to fully understand the degradation process, mechanism and influencing factor of MPs, as well as their environmental behavior. The main contents of this review are that: (1) Different degradation processes and mechanisms of MPs are

discussed; (2) The degradation pathway of MPs are analyzed; (3) The additives and DOM leached from MPs during the degradation process and their toxic effects on environment are elucidated; (4) The main factors affecting the degradation process are summarized. Finally, knowledge gaps and future research priorities based on current understanding are presented.

2. Degradation process of MPs

2.1. Physical degradation

MPs will undergo physical degradation in the natural environment. The wear process is related to the temperature, biological friction, freeze-thaw cycles and the abrasive forces of water, wind and particulate matter (e.g., sand and gravel). The degree of physical wear is reflected by the changes in surface roughness, size and color. Pristine MPs typically have smooth surfaces, whereas degraded MPs exhibit increased roughness and more granular areas (Luo et al., 2020c). On the other hand, an increased irradiation time results in a concomitant decrease in the average softening temperature of MPs and MPs become harder after photodegradation (Luo et al., 2021a).

Researchers used a low-energy rolling compression wet grinder to simulate the physical degradation of polystyrene (PS). In addition to the obvious reduction in particle size, there are also severe shearing phenomena after grinding. MPs may undergo cracking, detachment, and elongation in a single direction (Ravishankar et al., 2018). Others have investigated how low-density polyethylene (LDPE) breaks down when exposed to typical environmental conditions (temperature, wind, sand, etc.). The experiments were performed by conducting wind-blown sand simulation tests at 40 °C. Airflow is provided from the wind turbine at a pressure of about 100 kN/m^2 and a flow rate of 2.3 m^3/min . Sand particles randomly impact the sample surface in the way of “Brownian motion” in a closed chamber with the airflow. Windy sand causes frost on the surface of the film and forms micro-cracks, which results in irreversible damage to the polymer film. Small particles of sand embedded in the surface can also cause a significant reduction in ultraviolet-visible (UV-visible) light transmittance (Djakhdane et al., 2016).

Plastic will turn color after being exposed to UV light (Kowalski et al., 2016). Studies simulated the mechanical weathering of LDPE and polyethylene terephthalate (PET) in silt/sand and rainwater, respectively. The results demonstrated that both LDPE and PET MPs showed light yellow discoloration, though the discoloration was less pronounced in PET. Silt and sand accumulated on the surface of MPs and affected the transport behavior of pollutants (Aghilinasrollahabadi et al., 2021).

Physical wear of MPs can also cause the release of additives (He et al., 2023b; Lu et al., 2023), but there are very few studies on this subject. Most of the research on physical degradation examines the impact of sand, gravel and weathering. There are few studies on the effects of water and temperature on physical degradation, and thus the physical degradation of MPs in water needs further research.

2.2. Chemical degradation

The chemical degradation of MPs in natural waters is mainly driven by photoinduced reactive oxygen species (ROS) (Sun et al., 2020). The mechanism is similar to that of artificial reinforcement methods (e.g., AOPs, electrochemical and photocatalytic degradation) in the laboratory, but the rate is significantly different. This chapter focuses on natural photodegradation and the role of DOM, sulfides and reducing agents. The artificial degradation method is only discussed as a comparison. The formation of oxygen-containing groups during degradation significantly enhances the hydrophilicity, charge of MPs and surface polarity (Luo et al., 2022c). To investigate the physicochemical properties of MPs before and after degradation, simulation experiments are conducted in the laboratory. So far, chemical degradation methods

commonly used in the laboratory include electrochemical degradation, AOPs, photodegradation, photocatalytic degradation, etc (Fig. 1).

2.2.1. Photodegradation

Photodegradation is the destruction process of polymer by absorbing light, including the absorption of visible light, infrared rays and UV light (Bacha et al., 2021). This method uses the energy of light to excite chemical bonds within MPs molecules, facilitating their fragmentation into smaller molecules and eventually into harmless substances.

Exposure to UV-visible light triggers the generation of ROS (such as $\bullet\text{OH}$, $\text{O}_2^{\bullet-}$, $^1\text{O}_2$, etc.), which further leads to the formation of oxygen-containing functional groups and new polymers (Duan et al., 2022). ROS play an important role in the degradation process of MPs as shown in Fig. 2. The order of redox potential of three common ROS is $\text{SO}_4^{\bullet-}$ (2.5–3.1 V vs NHE) > $\bullet\text{OH}$ (1.9–2.7 V vs NHE) > $^1\text{O}_2$ (0.81 V vs NHE) (Xie et al., 2022). These high-energy molecular oxygen can attack the polymer chains of MPs, causing chain breakage and the degradation of MPs. For example, $\bullet\text{OH}$ has a high oxidation potential and therefore has a strong oxidation capacity, which contributes significantly to the degradation of MPs with light irradiation (Xiao et al., 2017). In some photochemical experiments, it has been found that simulated sunlight can significantly change the particle size and functional group of PS as well as the environmentally persistent free radicals (EPFRs) in water. The PS surface accumulates a lot of $\text{O}_2^{\bullet-}$ and $\bullet\text{OH}$ radicals (Zhu et al., 2020a). The water-phase microscopic observations of different kinds of MPs under 254 nm UV light have been studied. It was found that the number of particles with minimum size increased with time, with the smallest particles reaching a size of 74–177 μm after 24, 48 and 72 h of exposure (Hebner and Maurer-Jones, 2020). Photodegradation often leads to a significant decrease in molecular weight, as shown by gel permeation chromatography (GPC) measurements, and can be accompanied by up to 66 % mass loss under prolonged UV exposure (Shi et al., 2021). Under continuous light irradiation, MPs showed an obvious tendency to break into smaller fragments, which results in a significant reduction of polymer molecular weight. As a result of ROS, bisphenol A

(BPA) and other low molecular intermediates released from polymer chains can be trapped in MP particles or released into leachate to form dissolved organic matter (DOM) (Shi et al., 2021). It is suggested to further explore the mechanism and intermediate products in the photodegradation process. Many researchers have concentrated on studying the hazards associated with MPs and their leachate (Ding et al., 2022b; Gunaalan et al., 2020), but the volatile organic compounds (VOCs) released by MPs are often underestimated. Some studies propose the possible mechanism of VOCs released from MPs, and verify the effectiveness of these mechanisms (Biale et al., 2022; Lomonaco et al., 2020). Toxic compounds such as benzene, toluene and phenol may be released through MPs. The concentration of these releases may increase significantly, especially under UV irradiation. As the irradiation time becomes longer, the concentration of VOCs will be higher (Wu et al., 2022). So far, the research and development of photodegradation technology is still immature. For instance, the photodegradation process may take a long time to completely decompose MPs (Liu et al., 2022b), and its efficiency is affected by light conditions, polymer types and environmental factors (Zhang et al., 2022b). Additionally, the intermediates in the photodegradation process may have unknown effects on the environment (Li et al., 2022). Overall, the photodegradation technology of MPs has its potential.

The photodegradation of MPs is a very slow process in the natural environment, so we can introduce some catalysts during the photodegradation process to improve the efficiency of photodegradation.

2.2.2. Degradation of MPs by DOM, sulfides and reducing agents

In addition to the above four chemical degradation methods, DOM can also be applied to the degradation process of MPs. DOM refers to organic matter dissolved in water, such as humus, protein, carbohydrates, etc. The adsorption between MPs and DOM can change the surface properties of MPs, thus affecting their environmental behavior in water. This change in surface properties may affect the aggregation, deposition and dispersion patterns of MPs and their interactions with other pollutants or organisms (Zhu et al., 2020b). It is reported that

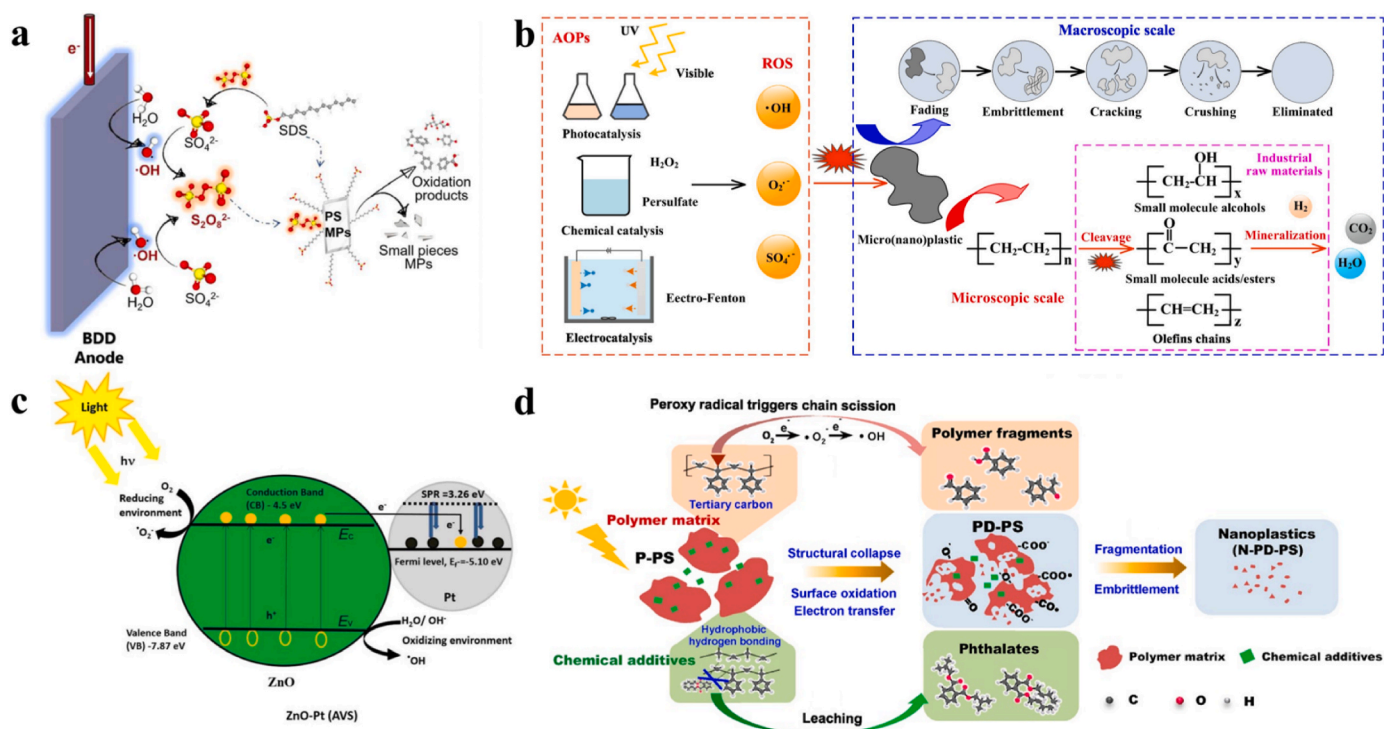


Fig. 1. Chemical degradation mechanisms of MPs. (a) Electrochemical degradation (Lu et al., 2022), Copyright 2022 Elsevier Ltd; (b) AOPs (Shen et al., 2022), Copyright 2022 Elsevier BV; (c) Photocatalytic degradation (Tofa et al., 2019b); (d) Photodegradation (Wang et al., 2020b), Copyright 2020, American Chemical Society.



DOM exhibits high binding affinity to PS in the order of fly ash > standard humic acid (SHA) > fluka humic acid (FHA, derived from natural coal). These differences may be due to the molecular structure of DOM, the presence of functional groups and surface charge (Ding et al., 2022a). This phenomenon has attracted more attention from researchers who aim to reveal the degradation process and mechanism between DOM and MPs. It was found that the interaction between MPs and DOM not only increased the surface roughness, but also accelerated the oxidation process. These changes further enhance the spectroscopic characteristic signals of MPs, making them easier to detect in environmental monitoring (Luo et al., 2022b). The introduction of fulvic acid

significantly increased the amount of oxygen-containing functional groups on the surface of MPs (Luo et al., 2024). In the degradation system, O_2^- , 1O_2 and EPFRs act together, significantly affecting the degradation process of MPs (Ouyang et al., 2023). Even in the dark, the presence of DOM leads to MPs degradation. This is because the ROS produced by DOM in the dark environment can still promote the oxidation process of MPs (Qiu et al., 2022). Studies have shown that natural DOM contains a higher proportion of aromatic hydrocarbons and achieves a better improvement effect in the degradation process of MPs compared to DOM derived from biochar. This improvement is manifested by a reduction in the particle size of MPs, the increase of

crystallinity, the decrease of polymerization degree, the destruction of surface morphology, and the increase in surface oxygen-containing functional groups (He et al., 2023a). Overall, the interaction between MPs and DOM is a complex and important research area, essential for comprehending the environmental behavior and toxic effects of MPs in water bodies.

Sulfide is a common nucleophile and reducing agent. Some researchers speculate that sulfide may also cause the oxidation of MPs. Specifically, sulfide participates in free radical reactions to generate active sulfide free radicals, which will oxidize MPs. Sulfides spontaneously oxidize to form $\bullet\text{OH}$ radicals, $\bullet\text{OH}$ act as major oxidants attacking carbon atoms in polymer chains, resulting in surface oxidation and chain breakage (Zhao et al., 2021). In addition, sulfide may also participate in redox reactions and react with oxidants to further promote the oxidation of MPs. For example, hydrogen sulfide (H_2S) and H_2O_2 react at a neutral pH through the Fenton mechanism to produce $\bullet\text{OH}$, which results in the oxidation of MPs (Liu et al., 2021b). Studies have proven that sulfide treatment of MPs causes physical damage and chemical reaction of MPs. The manifestation of physical damage in the MPs was observed to occur in the form of crack formation, increased surface roughness, reduced particle size, and increased specific surface area. The oxidation of sulfide in aqueous solution triggers the formation of ROS and thus degrades MPs (Lombardo et al., 2020). Sulfide addition reactions may cooperate with free radical oxidation to affect the conversion of MPs (Zhao et al., 2021). The presence of sulfide forms highly reactive and reducing thiol groups, which significantly affects MPs binding to heavy metal cations and oxygen anions (Duan et al., 2023). Overall, sulfides may affect the degradation processes of MPs by catalyzing oxidation reactions and changing the chemical properties of MPs. However, the specific influencing mechanism may be controlled by multiple factors, such as environmental conditions, types of MPs and sulfide concentration.

It was found that the addition of reducing agents (NaBH_4 , Na_2MoO_4 and HCl) could promote the rapid oxidation of MPs under dark and hypoxic conditions. The results showed that the particle size, crystallinity and effective ecological concentration of PET decreased and C/O ratio increased. When the three reducing agents were added at the same time, the damage of MPs was the most obvious and the change of surface morphology was the greatest (Xiao et al., 2023). However, this experiment used a relatively rare reducing agent. Other researchers selected four common reducing agents (NaBH_4 , ascorbic acid, Na_2S and $\text{Na}_2\text{C}_2\text{O}_4$) to construct reducing conditions and investigated the degradation process of PVC in the reducing environment. The reducing agents caused dechlorination of PVC through nucleophilic substitution reaction and the oxygen-containing functional groups of MPs were then attacked. The findings showed that PVC was degraded under four reducing conditions. Among them, the most pronounced degradation of PVC was observed in the presence of NaBH_4 (Zhang et al., 2023b). Reducing agents can enhance the degradation of MPs through various mechanisms, but the enhancing effect depends on factors such as the types of MPs and concentration of the reducing agents.

2.2.3. Artificial chemical degradation

Under laboratory conditions, researchers often use artificial enhanced degradation techniques (such as electrochemical degradation, AOPs and photocatalytic degradation) to verify similar reaction mechanisms under natural conditions. These techniques produce high concentrations of ROS (e.g., $\bullet\text{OH}$, $\text{O}_2^{\bullet-}$, $\text{SO}_4^{\bullet-}$, etc.) by applying electric fields, catalysts or oxidants, which mineralize MPs quickly and efficiently.

Photocatalytic degradation is a technology that uses light energy to decompose pollutants into harmless substances. Different from natural photodegradation, photocatalytic degradation requires external energy input. The fundamental principle is the absorption of energy from visible light or UV light through photocatalysts, which are typically semiconductor materials. Following the excitation of the catalyst, charge separation occurs which leads to the transition of free electron from valence band (VB) to conduction band (CB). Free electrons are excited to

form holes. ROS is produced when free electrons and holes react with H_2O , OH^- and O_2 adsorbed on the semiconductor surface (such as $\bullet\text{OH}$ and peroxides) (Uheida et al., 2021). The generated free electrons have strong reducing properties and can reduce water to hydrogen (Beltrán et al., 2019). The holes react with water molecules to generate free radicals such as $\bullet\text{OH}$ (Nabi et al., 2021). During the oxidation process, it is possible to eventually decompose MPs into carbon dioxide and water (Turchi and Ollis, 1990). In the process of MPs photocatalytic degradation, the photocatalyst will produce high-energy electrons and holes when it receives photons with energy higher than the band gap. Typically, light-induced electrons react with oxygen to form ROS, which in turn reacts to produce H_2O_2 and $\bullet\text{OH}$. Moreover, the holes can also react with OH^- or H_2O in the reaction system to form $\bullet\text{OH}$ (Bacha et al., 2021).

Photocatalysts initiate photochemical reactions under light irradiation and they are mainly divided into bare photocatalysts and composite photocatalysts. Compared with bare photocatalysts, composite photocatalysts have superior performance (Ge et al., 2022). TiO_2 is a bare photocatalyst. Because of the limited energy bandwidth of anatase TiO_2 , it mainly absorbs UV light at a wavelength of nearly 388 nm (Ge et al., 2022). To overcome this limitation, the researchers modified the catalyst. Researchers have found that a higher percentage of mass loss was observed in the degradation of high density polyethylene (HDPE) films using N- TiO_2 composite photocatalyst compared with the bare photocatalyst TiO_2 (Ariza-Tarazona et al., 2019). Currently, photocatalytic degradation still faces many challenges, such as slow degradation rate, poor photocatalyst stability and difficulty in catalyst recycling.

Electrochemical degradation is a technology that utilizes electro-oxidation to degrade and mineralize refractory organic substances, particularly persistent and toxic pollutants. It offers benefits such as environmental friendliness, ease of implementation, high efficiency and broad degradation capabilities (Chen et al., 2019). Electrochemical degradation technology enhances the formation of $\bullet\text{OH}$ and accelerates chain scission. The technique was used to verify the chain reaction pathway of ROS in natural photodegradation. During the electro-degradation process, nanoplastics (NPs) are mainly oxidized by two types of ROS on the surface of boron-doped diamond (i.e., $\bullet\text{OH}$ and $\text{SO}_4^{\bullet-}$). $\bullet\text{OH}$ is formed at the anode after water oxidation in the process of electrodegradation, while $\text{SO}_4^{\bullet-}$ is formed by activating $\text{S}_2\text{O}_8^{2-}$ through H_2O_2 . $\bullet\text{OH}$ reacts with the carbon-carbon double bond ($-\text{C}=\text{C}-$) of MPs and attacks the aromatic nucleus of MPs. $\text{SO}_4^{\bullet-}$ can attack a variety of aromatic and ring-like organic molecules (Kiendrebego et al., 2022). Electrochemical oxidation is categorized into anodizing and indirect cathodic oxidation. Anodic oxidation (AO) refers to the direct oxidation of organic contaminants on the anode surface through charge transfer, or the indirect oxidation in an aqueous solution through $\bullet\text{OH}$ or reagents (e.g., H_2O_2 and peroxymonosulfate) (Chen et al., 2022b). Indirect cathodic oxidation is an electro-Fenton (EF) technique. Fe^{2+} catalyzes the decomposition of H_2O_2 , leading to the production of $\bullet\text{OH}$ and other active oxygen radicals. These radicals play a crucial role in oxidizing MPs (Du et al., 2021). There is also an electro-Fenton-like process that degrades MPs through cathodic reductive dechlorination and $\bullet\text{OH}$ -driven oxidation (Chen et al., 2022b). It has been shown that the ROS generated by the electro-oxidation (EO) process, which employs in-situ generation of H_2O_2 , resulted in approximately a 2.6-fold higher degradation of NPs compared to the conventional EO process. The EO- H_2O_2 process achieved a degradation efficiency of $86.8\% \pm 1.8\%$ for NPs within 40 min at the optimized current density of 36 mA cm^{-2} (Kiendrebego et al., 2022). The electrochemical degradation technology has the potential to be a valuable tool in the field of MPs pollution control. However, further research and development are needed to ensure its effective application in the degradation and removal of environmental MPs.

AOP is a promising technology that provides an effective solution to the problem of MPs pollution. MPs can be degraded and converted into harmless substances by highly oxidizing active species, thus reducing

environmental pollution. In contrast to photodegradation, AOPs are forced to produce high concentrations of ROS by the addition of oxidants and catalysts. For instance, hydrogen peroxide, persulfate and ozone are all oxidizing substances commonly used to degrade MPs in advanced oxidation. In AOPs, various reactive oxygen species with high oxidation potential, including $\bullet\text{OH}$, superoxide ($\text{O}_2^{\bullet-}$) and sulfate radicals ($\text{SO}_4^{\bullet-}$), play a significant role in the degradation of MPs (Kang et al., 2019). MPs can even be completely mineralized after undergoing intensive oxidation by AOPs (Dos Santos et al., 2023).

Fenton reaction catalytically decomposes H_2O_2 to produce $\bullet\text{OH}$ radicals in the presence of Fe^{2+} (Hu et al., 2022b). In a hydrothermal coupled Fenton system, polyethylene (PE) lost 9.16 % of weight after 95 h and the mineralization efficiency reached 6.12 % after 75 h (Hu et al., 2022a). During the process of MPs being treated with persulfate, various free radicals (e.g., $\bullet\text{OH}$ and $\text{O}_2^{\bullet-}$) are generated to oxidize MPs (Ren et al., 2021). During the degradation process of heat-activated $\text{K}_2\text{S}_2\text{O}_8$, the originally smooth outer surface of MPs undergoes a transformation, becomes rough, cracked and pitted. This transformation enhances the hydrophilicity of the degraded MPs due to increased oxygen-containing functional groups (Kong et al., 2021). The synergistic effect of ozonation and H_2O_2 synergistically enhances PE degradation. Under conditions of the O_3 flow rate of 3 L/min and H_2O_2 concentration of 1 mol/L, this approach significantly intensify O-H and C=O bands, resulting in a carbonyl index value of 1.33. AOPs have the potential to significantly reduce the degradation period of MPs, as well as to decompose complex and persistent pollutants (Sutkar et al., 2023). Furthermore, AOPs showed excellent performance in degrading the pigment released by MPs (Luo et al., 2021b).

2.3. Biological degradation

In natural ecosystems, microorganisms, fungi and invertebrates are continuously involved in the degradation of MPs through biocatalysis. This process depends on the synergistic effect of natural enzymes secreted by organisms and environmental conditions, and constitutes a key link in the Earth's material cycle (Fig. 3). Microorganisms are one of the main driving forces for MPs biodegradation. Microorganisms secrete a variety of enzymes such as esterases, ureases, lipases, proteases, glycoside hydrolases, and laccases. Different enzymes for the degradation of different types of MPs are presented in Table 1. Enzymes secreted by microorganisms are divided into extracellular enzymes and intracellular enzymes. Due to the high molecular weight of polymer, the utilization and absorption of MPs by microorganisms are challenging. Hence, both intracellular and extracellular enzymes are crucial in

breaking down the bonds of MPs. Microorganisms secrete extracellular enzymes that hydrolyze long-chain polymers into monomers and enter the interior of microbial cells, where they are subsequently metabolized to CO_2 and H_2O by intracellular enzymes. This process provides an energy source for microorganisms (Zhang et al., 2021b). In the process of enzymatic hydrolysis, MPs undergo the process of surface erosion. Because of the uneven erosion rate, the surface of the MPs becomes rough, forming ridges, branches and rods (Wei et al., 2022).

Studies demonstrated that certain bacteria (*Ideonella sakaiensis*) are capable of hydrolysing PET through the secretion of two specific enzymes that facilitate the breakdown of PET into simpler compounds. These breakdown products are then further utilized by the bacteria as basic building blocks for growth, enabling the biodegradation of plastics (Yoshida et al., 2016). Several studies have explored how fungi can degrade MPs and revealed that fungi, like bacteria, can use MPs as carbon sources (Yuan et al., 2020). Different fungal strains were used to biodegrade LDPE. It was found that *Aspergillus versicolor* strain is a more potential candidate for LDPE degradation than *Aspergillus flavus* and *Fusarium solani*. It is anticipated that this will provide a solution to the problem of MPs pollution (Das et al., 2018). Because algae can uptake pollutants in the environment and metabolize them into harmless components, some researchers use algae to degrade MPs in water. Microalgae may utilize the toxins or enzymes to degrade plastic polymers (Abdelfattah et al., 2022). Researchers have suggested that algae such as *Chlamydomonas reinhardtii*, *Conus algae* and *Triatomycetes* could produce an enzyme (PETase) to degrade PET (Karalija et al., 2022). Studies show that certain insects can also degrade MPs. For example, the super worm (*Zophobas atratus*) uses polystyrene foam (PS foam) as its only food. The depolymerization of PS occurs in the intestine of the superworm, and up to 36.7 % of PS is mineralized into carbon dioxide (Yang et al., 2020).

Some researchers directly use enzymes to degrade MPs. However, enzymes have extremely high requirements for the working environment. Inappropriate (too high or too low) temperature and pH will affect the working efficiency of enzymes, and even change their structure to make them inactive. Due to the substrate specificity of enzymes, different types of MPs require different enzymes for biodegradation. Laccase and alkane hydrolase have a significant degradation effect on PE, and hydroquinone peroxidase from *Azotobacter nortis* HM121 can degrade PS (Othman et al., 2021). Isolates such as *Ideonella sakaiensis* 201-F6 produce PETase, which is capable of degrading PET to terephthalic acid and ethylene glycol (Anand et al., 2023). In order to develop more efficient bioremediation methods, some researchers have used modern biotechnology to modify the genes of microorganisms, plants

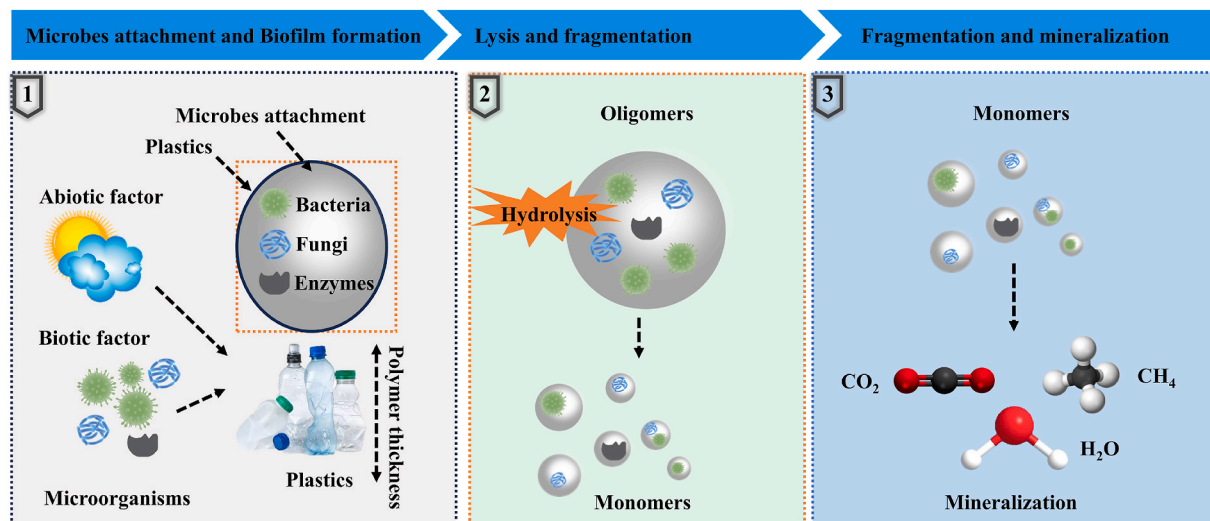


Fig. 3. Biodegradation mechanisms of MPs. Adapted from reference (Bacha et al., 2021).

Table 1
Degradation of MPs by different bioenzymes.

MPs type	Typical species	Source	Duration of degradation (d)	Weight loss (%)	Reference
PE	Laccase	<i>E. coli</i>	6	84	Zhang et al. (2023a)
	Laccase-like multicopper oxidases (LMCOs)	<i>Aspergillus flavus</i> PEDX3	28	3.90 ± 1.18	Zhang et al. (2020)
PET	BIND-PETase	<i>E. coli</i>	7	7	Zhu et al. (2022a)
	Leaf-branch compost cutinase (LCC)	Plant compost metagenome	5	47	Tiso et al. (2021)
	Glycoside hydrolases	<i>Ideonella sakaiensis</i> 201-F6	60	–	Yoshida et al. (2016)
PS	thiol peroxidase (<i>tpx</i>)	<i>Enterobacter hormaechei</i> LG ₃	30	3.82 ± 0.77	Kang et al. (2023)
	Alkyl Hydroperoxide reductase C (<i>ahpC</i>)				
	Bacterioferritin comigratory protein (<i>bcp</i>)				
PUR	Serine hydrolase (SH)	<i>Pseudomonas</i> sp	15	2.6	Kim et al. (2020)
	Esterase	<i>Bacillus subtilis</i> MZA-75	28	–	Shah et al. (2013)
PVC	LC cutinase (LCC)	Plant compost metagenome	8	4.9	Schmidt et al. (2017)
	Lignin peroxidase	<i>Phanerocheate chrysosporium</i>	30	22	Khatoun et al. (2019)

and animals. For example, a computer-guided mutagenesis technique was used to replace six specific amino acids in *Archaeoglobus* carboxylesterase with those in *Fusarium* BTA hydrolase. This conversion enhanced the ability of the carboxylesterase to degrade PET (Lameh et al., 2022). Other researchers have combined biological and non-biological degradation methods to achieve more efficient degradation of MPs. They design a bio-non-bio photocatalytic system by assembling *Methanobacterium* and carbon dot-functionalized polymeric carbon nitride, which converts thermally pretreated biodegradable MPs into CH₄ with a conversion rate close to 100 % (Ye et al., 2022). Therefore, microbial enzymes play a key role in environmental restoration and ecological health through biodegradation, and they make the biodegradation of MPs more competitive, clean and safe by the advantages of high efficiency and functionality.

Although some progress has been made in the biodegradation process of MPs, further research is still needed to gain a deeper understanding of the degradation mechanism. It is important to recognize that relying solely on biodegradation for addressing MPs pollution is not sufficient. Instead, it should be employed in conjunction with other approaches, such as reducing plastic use and increasing recycling rate, in order to address this issue comprehensively.

3. Degradation pathway of different MPs

The polymer's chemical bonds are broken and MPs are degraded into small molecules. These molecules can then be transformed into useful products or fully mineralized into CO₂ and H₂O. This process, called MPs

degradation, may be induced by physical, chemical and biological processes (Zhang et al., 2021b). The main mechanism of MPs degradation is hydrolysis and oxidation. Taken together, the MPs degradation is a complex process involving the interaction of multiple biotic and abiotic factors. A thorough understanding of these degradation pathways is of great importance for the development of efficient MPs degradation methods.

3.1. Degradation pathway of PE

The pathways of chemical degradation and biological degradation of PE are shown in Fig. 4. It is commonly accepted that the excitation of photocatalysts at an optimal light energy level results in the formation of •OH radicals, which possess a high oxidizing ability and are capable of degrading organic pollutants. Therefore, •OH play an important role in the photodegradation of MPs. •OH and O₂^{•−} generated by the catalyst induce the degradation of the long-chain PE polymer, such as attacking the chromophoric groups and defects. This process results in the production of alkyl radicals, which subsequently undergo chain scission, branching and cross-linking (Vital-Grappin et al., 2021). Then, oxygen interacts with the polymer chain, leading to the formation of peroxy radicals and the release of hydrogen atoms, resulting in the formation of H₂O₂ groups. H₂O₂ molecules are the most significant oxidation products, regulating the rate of photocatalytic degradation of polymers. The alkyl radicals break down into alkoxy (RO•) radicals and in subsequent reactions produce carbonyl (C=O) and vinyl (•HC = CH₂) radicals, as well as •OH radicals. All these radicals play a major role in the chain

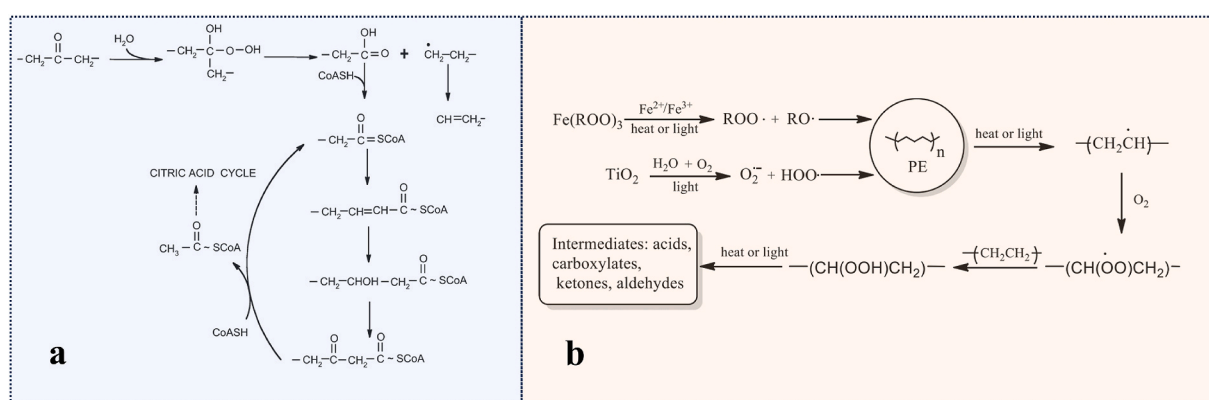


Fig. 4. Schematic diagram of degradation path of PE-MPs. (a) Proposed biodegradation mechanism of PE. One proposed mechanism of biodegradation of PE polymer that has been altered by the addition of carbonyl (C=O) group in its polymer chain (Gautam et al., 2007), Copyright 2007, Humana Press Inc; (b) Chemical degradation. The free radicals are generated on the polymer chain, due to the action of heat and/or light (Fa et al., 2020), Copyright 2019, Springer-Verlag GmbH Germany, part of Springer Nature.

cleavage (Bacha et al., 2021; Tofa et al., 2019a).

PE is not biodegradable due to its high hydrophobicity and high molecular weight (Krupp and Jewell, 1992). The biodegradability of PE can be realized by improving the hydrophilicity of PE and shortening the length of polymer chain (Bikiaris and Karayannidis, 1999). The combined effect of abiotic and biotic processes results in the fragmentation of the bulk polymer, thereby increasing the biodegradable surface area. Some extracellular enzymes may further cleave the polymer. After the polymer is initially cleaved into oligomers of 10–50 carbon atoms, it will be transported into cells for subsequent metabolism and eventually mineralized into CO_2 , H_2O , CH_4 , etc (Lucas et al., 2008; Sivan, 2011).

3.2. Degradation pathway of PP

PP degrades under high temperature or radiation due to the susceptibility of its tertiary hydrogen atoms to oxygen attack (Calvert et al., 2011). The degradation pathway of PP is shown in Fig. 5. Photodegradation of PP is caused by various impurities. These impurities and extraneous groups in PP absorb UV light, leading to the formation of alkyl radicals by transferring hydrogen within the chain (Ranby, 1989). The chemical structure of PP consists of C-C and C-H bonds. In comparison to ester bonds, single bonds are observed to be more stable in the process of degradation. Additionally, enzymes secreted by various microorganisms have the capability to oxidize C-C and C-H in polymer chains. Enzymes such as alkane hydroxylase, alcohol dehydrogenase and aldehyde dehydrogenase attach to the surface of PP and then trigger the decomposition process. Moreover, enzymes facilitate the conversion of carbonyl groups into carboxylic acids and the breakdown of carbon chains into smaller hydrocarbons, leading to the release of alkanes and alkenes. Small molecule hydrocarbons are metabolized by microorganisms into CO_2 and H_2O (Zhang et al., 2022a).

3.3. Degradation pathway of PET

The degradation pathway of PET is shown in Fig. 6. The photodegradation of PET under near-ultraviolet irradiation results in chain breakage through Norrish I and II reaction. Cross-linking also occurs and

the polymer becomes brittle, discolored and has an uneven surface (Edge et al., 1991). When the polymer is exposed to air, it triggers a complex series of oxidation reactions. The alkyl radicals formed during photodegradation react with oxygen in the air to form $\text{ROO}\bullet$ radicals. These radicals further participate in the reaction and eventually produce CO_2 , which is also the main volatile product in the process of photodegradation (Day and Wiles, 1972).

It is found that the bacterium *lideonella sakaiensis* 201-F6 is capable of degrading and assimilating PET through a metabolic pathway. This bacterium is capable of secreting two enzymes (PETase and MHETase), which are responsible for the degradation of PET (Yoshida et al., 2016). PETase catalyzes the conversion of PET into mono (2-hydroxyethyl) terephthalate (MHET) and bis (2-hydroxyethyl) terephthalate, while MHETase facilitates the transformation of MHET into terephthalic acid and ethylene glycol (Thakur et al., 2023).

3.4. Degradation pathway of PS

The chemical and biological degradation pathways of PS MPs are shown in Fig. 7. The photodegradation of PS MPs is mainly initiated by UV light. The benzene ring structure of PS forms an excited state molecule after absorbing UV light in the 280–320 nm band. The photodegradation pathway of PS is primarily driven by the strong oxidizing and non-selective attack of $\bullet\text{OH}$ radicals. Studies have shown that the concentration of $\bullet\text{OH}$ radicals generated in the PS nanoparticle system is higher. Furthermore, $\bullet\text{OH}$ radical addition reactions preferentially occur in regions of high electron density (such as benzene rings), resulting in the formation of phenols and monohydroxylated derivatives (Xiao et al., 2014). Simultaneously, $\bullet\text{OH}$ induce the formation of tertiary polystyrene radicals through hydrogen abstraction and cleavage of the main chain C-C bonds. These radicals are further oxidized to $\text{RO}\bullet$ and generate hydroxylated intermediates in a chain reaction (Bacha et al., 2021). In subsequent reaction process, the $\bullet\text{OH}$ attack leads to the cleavage of C2-C3 bond, producing aliphatic ketones (1722 cm^{-1}) and acyl radicals ($\text{R-CO}\bullet$). Acyl radicals combine with $\text{RO}\bullet$ to form ester products (1731 cm^{-1}). Continued attack by $\bullet\text{OH}$ or $^1\text{O}_2$ may further lead to the ring opening of benzene ring, producing ring-opened byproducts

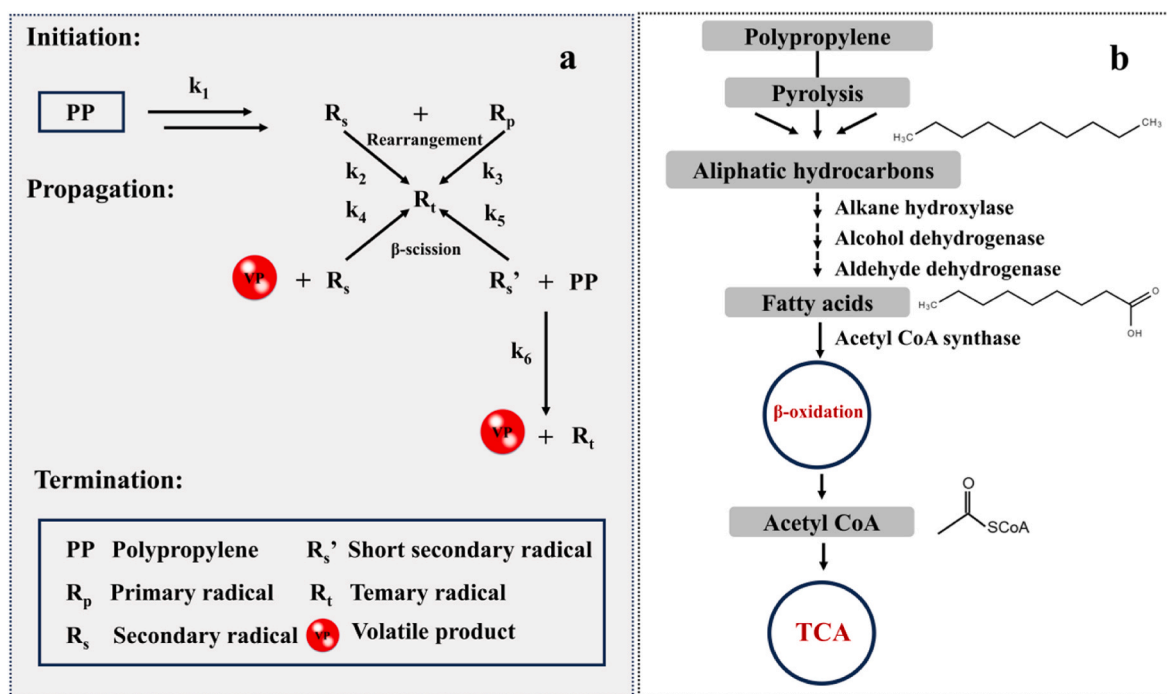


Fig. 5. Schematic diagram of degradation path of PP-MPs. (a) Chemical degradation; (b) Biodegradation. Adapted from reference (Ali et al., 2021; Bockhorn et al., 1999).

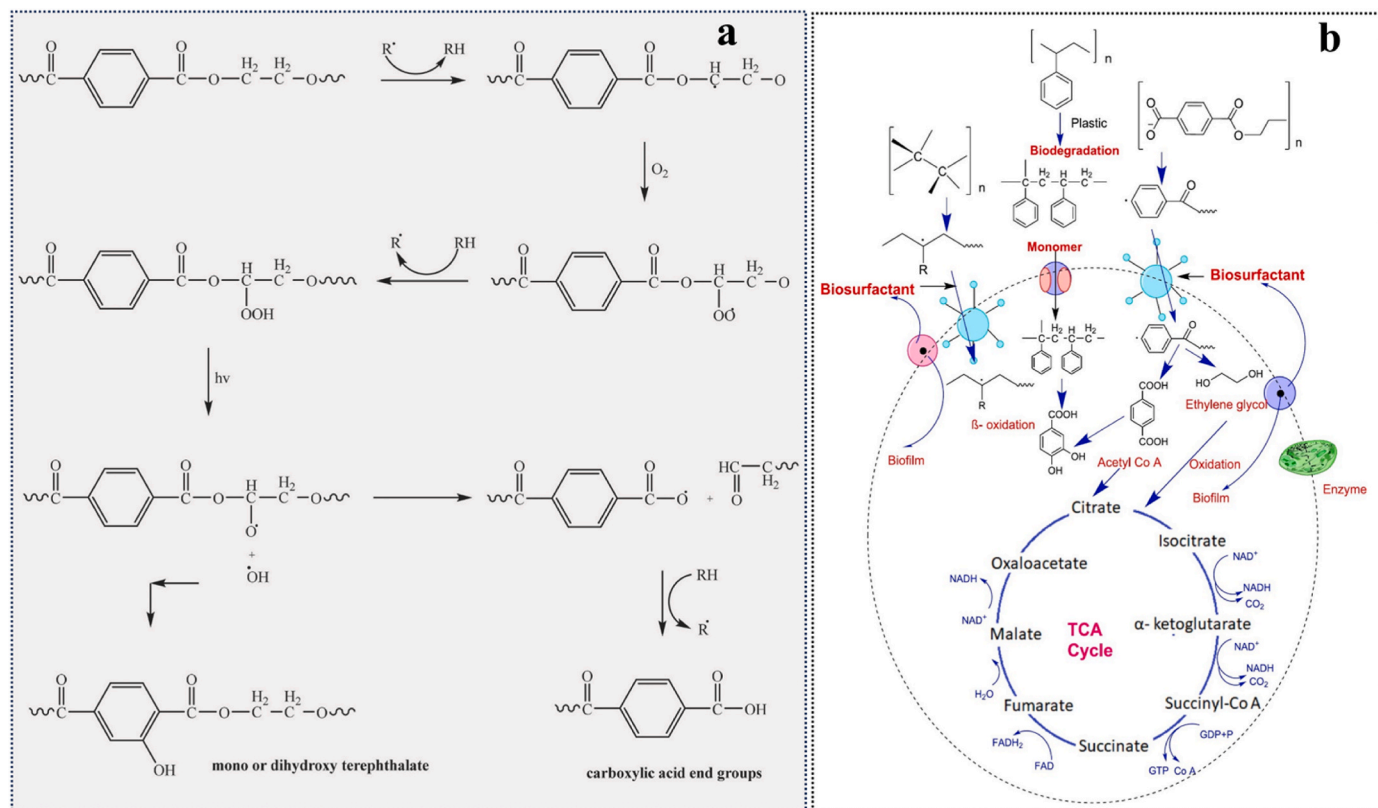


Fig. 6. Molecular mechanism of degradation of PET. (a) The mechanism for the photo-oxidation reaction undergone by PET when exposed to UV light (Horne et al., 2020); (b) Biodegradation pathways of PET (Thakur et al., 2023), Copyright 2023 Elsevier BV.

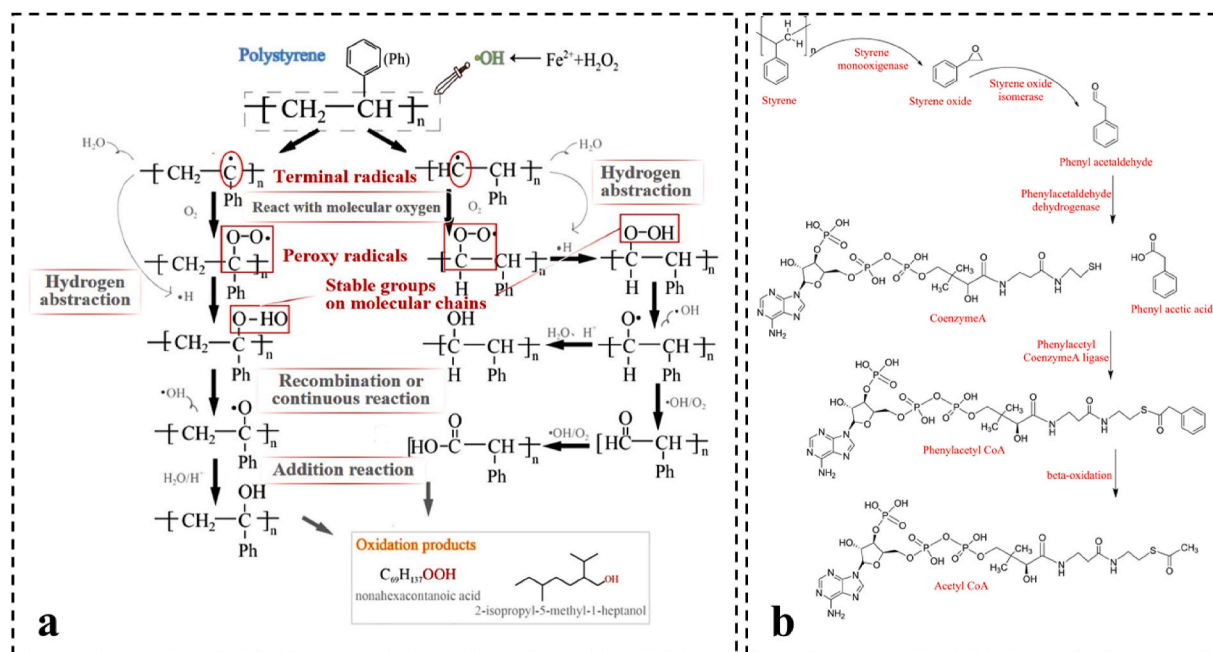


Fig. 7. (a) Proposed degradation pathway of PS (Yang et al., 2022), Copyright 2022 Elsevier Ltd; (b) Molecular mechanism of biodegradation of PS (Thakur et al., 2023), Copyright 2023 Elsevier BV.

such as aldehydes (Duan et al., 2022).

The biodegradation process of PS begins with styrene and depends on a series of microbial enzymatic reactions. First, styrene is oxidized to styrene oxide by the enzyme styrene monooxygenase. Styrene oxide is then isomerized to phenylacetaldehyde catalyzed by styrene oxide

isomerase. Phenylacetaldehyde is oxidized to phenylacetic acid under the action of phenylacetaldehyde dehydrogenase. In the intermediate metabolic stage, phenylacetic acid is conjugated with coenzyme A (CoA) under the catalytic action of phenylacetyl-CoA ligase to form phenylacetyl-CoA. Subsequently, phenylacetyl-CoA undergoes further

degradation through the β -oxidation pathway, ultimately converting into acetyl-CoA. Acetyl-CoA enters the tricarboxylic acid cycle for complete mineralization (Danso et al., 2019; Ho et al., 2018; Othman et al., 2021).

3.5. Degradation pathway of PVC

The degradation pathway of polyvinyl chloride (PVC) is shown in Fig. 8. Photooxidation degradation of PVC under the combined action of light and oxygen is a complex process, primarily proceeding through three pathways (Fig. 8a): (i) direct photo-induced dechlorination, (ii) oxygen-induced oxidation, and (iii) oxidation and dechlorination driven by $\bullet\text{OH}$ (Wang et al., 2020a). In Route (i), PVC initially undergoes C-Cl bond cleavage under light irradiation, leading to the generation of chlorine radicals and subsequent dechlorination reactions. This process results in the formation of carbonaceous intermediates (R1 and R2). Subsequently, these dechlorinated carbonaceous intermediates are further transformed under light irradiation into aging products, including olefinic (P1) and polyene (P2) structures (Christian and Michel, 1982; Jean-Luc and Jacques, 1991). In Route (ii), the introduction of oxygen initiates a peroxide chain reaction. The conjugated double bonds (P2) of PVC react with O_2 to generate cyclic peroxides (R3). Then, these peroxides decompose, producing free radicals (R4). The rearrangement of these free radicals leads to the formation of α - β unsaturated ketones and alcohols (P3), along with the generation of alkoxy radical intermediates (R5) (Decker and Balandier, 1981; Liu et al., 2019b). This oxygen-induced peroxide pathway significantly accelerates the photodegradation of PVC. In Route (iii), the presence of $\bullet\text{OH}$ plays a dominant role in the photodegradation of PVC. $\bullet\text{OH}$ can directly attack the olefinic groups (P3) in PVC chains, inducing oxidation and generating carbonyl-containing products (P4). Subsequent light irradiation and $\bullet\text{OH}$ attacks promote further dechlorination and oxidation reactions, resulting in the formation of new carbonaceous intermediates (R6). These intermediates undergo deprotonation and hydroxyl radical addition, leading to the formation of products (P5 and

P6). Among them, P6 can be further oxidized by $\bullet\text{OH}$ to generate polyketone-type products (P7). Simultaneously, P5 undergoes additional oxidation by $\bullet\text{OH}$, forming higher oxidation state products (P8) (Kim et al., 2017; Wang et al., 2020a).

Due to the lack of hydrolyzable ester bonds, biodegradation of PVC in the environment is particularly slow. However, studies have shown that microbial communities are able to mineralize PVC through a variety of enzymatic reactions (Fig. 8b) (El-Dash et al., 2023; Xu et al., 2023; Zhang et al., 2022c). Biodegradation of PVC typically begins with enzyme-catalyzed depolymerization reactions. In the presence of catalase and peroxidase, PVC polymers are partially depolymerized to lower molecular weight polymer intermediates. Then, these polymer intermediates are oxidized to hydroxyl-containing intermediates by the catalysis of monooxygenases and alkane hydroxylases, which are further oxidized to aldehydes by alcohol dehydrogenase and aldehyde dehydrogenases. In subsequent metabolic stages, enzymatic reactions such as monooxygenases, esterases and dioxygenases convert intermediates into fatty acids. Fatty acids enter the β -oxidation pathway to produce acetyl-CoA. Eventually, acetyl-CoA is completely oxidized to CO_2 and H_2O_2 by tricarboxylic acid cycle (the TCA cycle) (Khandare et al., 2021; Lv et al., 2024; Zhang et al., 2022c).

In general, the degradation process of MPs is complex and based on the specific characteristics of MPs and the prevailing environmental conditions. In order to effectively degrade MPs, a combination of multiple methods is usually required. With the deepening of research, more efficient and environmentally friendly degradation technologies of MPs may be developed.

4. Factors affecting MPs degradation

Factors influencing the MPs degradation process can be generally categorized into intrinsic characteristics of MPs and external environmental conditions (Fig. 9). The intrinsic characteristics can be broadly defined as polymer type, molecular weight, chemical stability, crystallinity and additives. External environmental conditions include

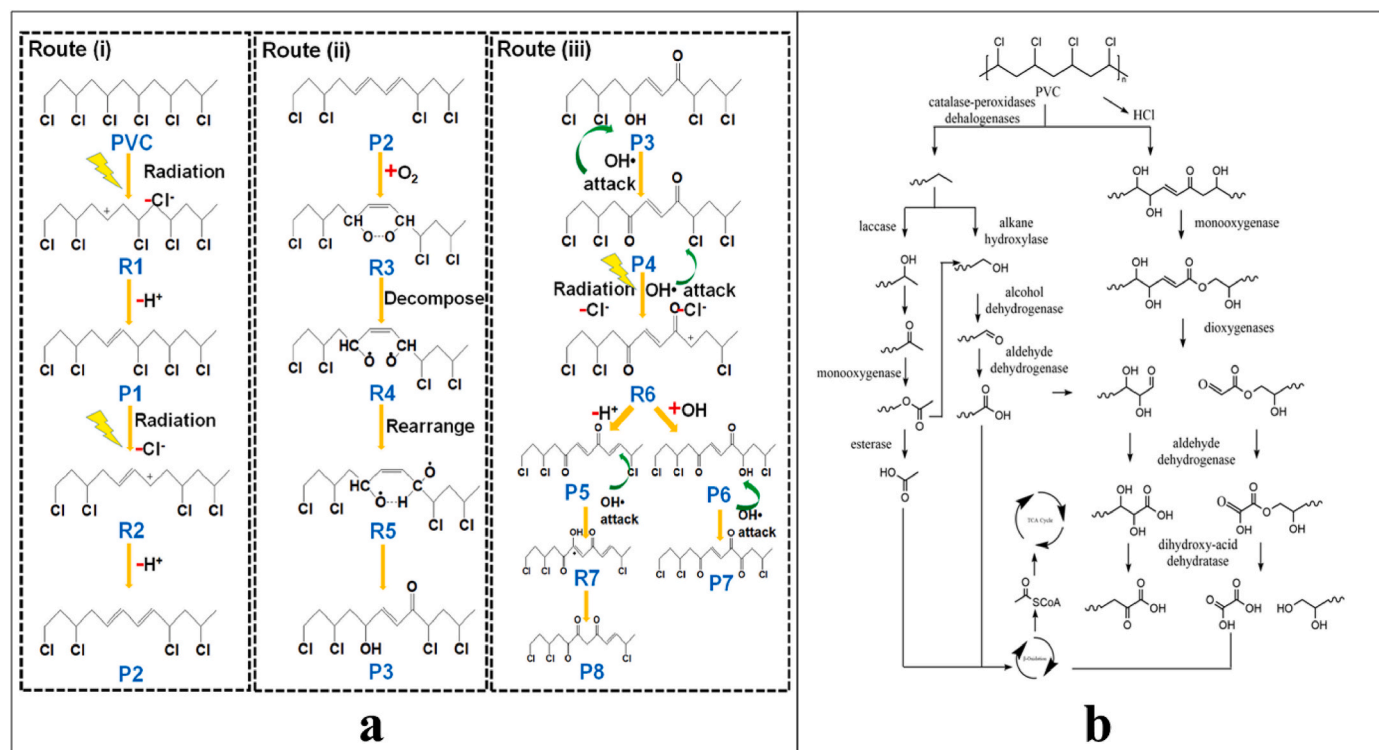


Fig. 8. (a) Illustration of PVC photodegradation routes: (i) Direct light radiated-dechlorination; (ii) O_2 -induced oxygenation of formed polyene; (iii) $\text{OH}\bullet$ -induced dechlorination and oxidation reactions (Wang et al., 2020a), Copyright 2020 Elsevier Ltd; (b) Biodegradation pathways of PVC plastics (Lv et al., 2024).

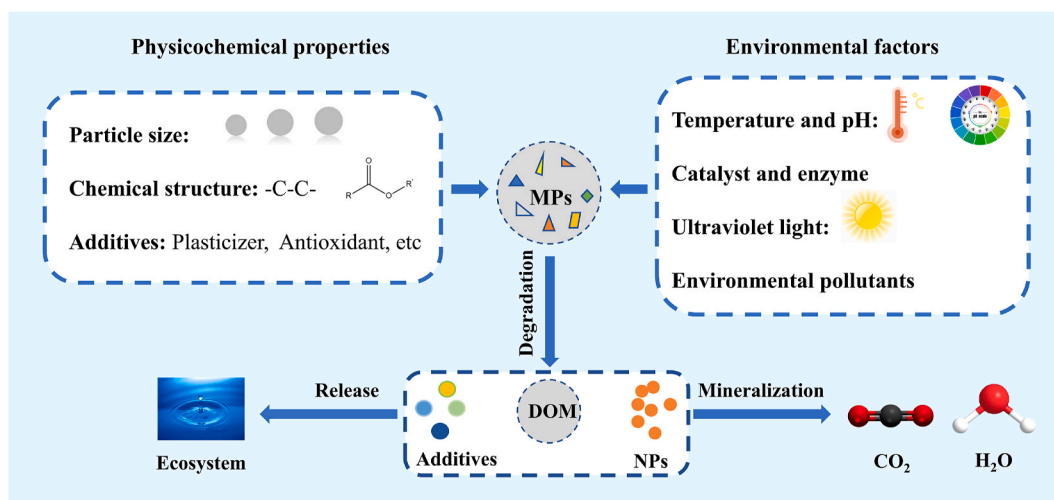


Fig. 9. Influencing factors of MPs degradation. MPs: Microplastics; NPs: Nanoplastics; DOM: Dissolved organic matter.

mechanical forces, friction, oxygen concentration, temperature, pH, microorganisms, etc. In the natural environment, the degradation of MPs takes a considerable amount of time, with the specific time period varying according to the impact of the aforementioned factors (Ge et al., 2023). These factors often do not act alone but interact with each other (Liu et al., 2022a). For instance, the adsorption mechanism between heavy metals and MPs is influenced by electrostatic adsorption, surface complexation and adsorption interactions (Luo et al., 2023).

4.1. The intrinsic properties of MPs

The intrinsic characteristics of plastic polymers significantly influence how quickly MPs degrade (Lambert et al., 2013). The degradability of MPs is influenced by their composition and structure. Furthermore, the presence of different forms of plastics also affects their degradability. The larger the specific surface area, the faster the degradation rate, since most plastic degradation occurs at the plastic surface (Gewert et al., 2015). However, the fragmentation of MPs leads to the adsorption of more pollutants that inhibit MPs degradation. For example, some pollutants can inhibit the photodegradation of MPs by absorbing or blocking UV radiation. Antibiotics and persistent organic pollutants can induce toxicity to the biofilm adhered to MPs surfaces, thereby hindering the biodegradation process (Zhang et al., 2021a). Different molecular compositions of plastics will result in changes in the hydrophobicity of plastics, thus affecting their adsorption behavior (Artham and Doble, 2008). According to previous reports, plastic polymers with ester bonds tend to biodegrade more readily compared to those lacking ester bonds (Barlow et al., 2020). The special structure and complexity of MPs composition will affect their degradation efficiency (Albertsson and Karlsson, 1993). Molecular weight directly determines the chain length and mechanical strength of MPs. Higher molecular weights are often associated with slower degradation rates (Nunes et al., 1982). The higher the crystallinity of MPs, the stronger their scratch resistance and wear resistance (Kanaga Karupiah et al., 2008). In other words, plastics with higher crystallinity are less prone to physical degradation. Traditional plastics like PE, PP, and PS exhibit minimal biodegradability in the natural environment due to the presence of highly stable carbon-carbon bonds and high molecular weight polymers in their main chains. The above factors make them chemically inert. Moreover, the absence of natural enzymes that can directly break down C-C bonds hinders the biodegradation of these olefin polymers (Taniguchi et al., 2019).

The color of MPs may also influence the photodegradation of MPs. In particular, black is more light-absorbing than light-colored, resulting in more significant photodegradation of black MPs (Jiang et al., 2021). The

use of different antioxidants and stabilizers can also slow down the degradation of plastics. Conversely, incorporating photosensitizer facilitates the photodegradation process of plastics. Consequently, the type of additives determines their effect on the degradation of MPs.

4.2. External environmental conditions

The principal external environmental factors influencing MPs degradation include temperature, light conditions, water quality, microorganisms, mechanical action, natural organic matter (NOM), and various cations and anions. Catalysts change the reaction rate (increase or decrease) without affecting the chemical balance (Datta et al., 2023). Hydroxyl-rich ultrathin BiOCl (BiOCl-x) shows great potential in photocatalytic degradation of MPs. The mass loss of plastics degraded by BiOCl-x was 24 times that of BiOCl nanosheets. The surface hydroxyl groups of BiOCl could effectively enhance the generation of $\bullet\text{OH}$, thereby improving the degradation performance (Jiang et al., 2021). Researchers studied the hydrolytic degradation of polylactic acid (PLA)/polyethylene oxide (PEO)/carbon nanotubes (CNTs) nanocomposites in neutral phosphate buffered saline solution. The results showed that CNTs promote the degradation of mixed PLA and PEO (Zare et al., 2019). Enzymes are also a type of catalyst. The enzymatic reaction produces carboxyl and hydroxyl groups, thereby enhancing the hydrophilicity and biodegradability of the polymer (Ru et al., 2020).

Low pH and low temperature have a comprehensive impact on the degradation of MPs (Liu et al., 2022a). In acidic conditions, the presence of H in the degradation system is beneficial to the degradation of MPs and the good interaction between MPs and colloidal nanoparticles. Low temperature causes MPs to break down and thus increases their specific surface area (Ariza-Tarazona et al., 2020). Acidic conditions may slow down the biodegradation of PE by *Pseudomonas aeruginosa*, but this inhibition could be offset at low temperatures (Tamnou et al., 2021). Some studies have found that increasing UV light intensity can significantly enhance ROS formation and photodegradation of MPs (Liu et al., 2021a). A higher degree of wear leads to increased shear and friction forces, which in turn results in greater fragmentation (Dai et al., 2023). UV light and mechanical abrasion play a more substantial role in the photooxidation and fragmentation of MPs. In an alkaline environment, the negative charge on the surface of catalyst and plastic particles leads to Coulomb repulsion, which reduces the adsorption and catalytic performance (Jiang et al., 2021). The oxygen content in the surrounding environment determines the degradation method of MPs. Biodegradable plastics can be fully transformed by aerobic microorganisms into CO_2 , H_2O , minerals and biomass within a limited time. Alternatively, the plastics are converted into CO_2 , methane and humic substances in the

anaerobic biodegradation process (Matjašič et al., 2021). In addition, the fungicide prothioconazole has been found to enhance the degradation of PE and biodegradable poly(butylene-adipate-co-terephthalate) plastic films (Li et al., 2020).

In addition to the abovementioned factors, the presence of NOM and various cations and anions in aquatic environments significantly influences the degradation of MPs. NOM includes humus and other DOM. It can be used as photosensitizer to accelerate the production of ROS and promote the photodegradation of MPs under light (Luo et al., 2022b). Cations such as Ca^{2+} , Mg^{2+} and Fe^{3+} , as well as anions such as Cl^- , SO_4^{2-} and NO_3^- , also play important roles in MPs degradation. For example, Fe^{3+} and NH_4^+ could inhibit the photoaging of PS-MPs by different mechanisms and the inhibitory effects increased with their concentrations. However, Ca^{2+} and Na^+ had no significant effect on PS-MPs photoaging due to their stable chemical property. Among them, Fe^{3+} inhibited the direct photodegradation of PS-MPs by weakening the light intensity through optical shadowing effect. NH_4^+ inhibited the indirect photooxidation of PS-MPs by $\bullet\text{OH}$ and decreased the concentration of $\bullet\text{OH}$ (Deng et al., 2024). Studies found that anions such as NO_3^- , Cl^- and Br^- promoted photoaging of PS-MPs. The ability of anions to promote photoaging was $\text{NO}_3^- > \text{Br}^- > \text{Cl}^-$. These anions affected the indirect photo-oxidative degradation of PS-MPs mainly by adjusting the concentration of $\bullet\text{OH}$. Halides not only regulated the activity of $\bullet\text{OH}$, but also reacted with aged PS-MPs or $\bullet\text{OH}$ to form halogenated organic compounds, which further participated in the photoaging process (Zhu et al., 2022b).

To reduce the persistence and environmental impact of MPs, it is important to understand these influencing factors and their interactions. Further research is required to investigate these factors in greater depth and to explore innovative degradation technologies and natural processes of MPs.

5. Release of additives and DOM during MPs degradation

Chemicals leached from plastics are usually plastic additives, and organics will produce when polymers are degraded (Romera-Castillo et al., 2018) (Table 2). Since additives easily leach from MPs into water, they have toxic effects on aquatic organisms. Moreover, as the degradation time of MPs increases, there is an elevated release of additives and the higher the toxicity to aquatic organisms (Luo et al., 2020a). UV irradiation induces the release of DOM from MPs and the DOM is a complex mixture of additives, oligomers, monomers and oxidation products (Lee et al., 2020). The DOM absorbs UV light, which can change the distribution and utilization of light energy. Therefore, the presence of DOM increases the possibility of forming free radicals on the surface of MPs (Wong et al., 2020).

5.1. Release of additives

Plastic additives include stabilizers, plasticizers, fillers, antioxidants, etc. In order to achieve the desired properties, plastic products require the addition of additives in relatively high concentrations, ranging from 10 % to 50 % (Oehlmann et al., 2009). The incorporation of these additives into the polymer alters its physicochemical properties, thereby enhancing its resilience to external factors such as light, heat and radiation. Most of these chemicals are non-biodegradable, they tend to accumulate in the environment through bioaccumulation pathways. Because the additives do not bind firmly to the carrier, they may be released from the plastic when MPs are decomposed into micron and nanometer sizes in the environment. Studies have demonstrated that the flame retardant diffuses from the MPs matrix, then through the MPs-medium boundary layer and finally into the environment (He et al., 2023b). Researchers studied how fluorescent additives leach from polyurethane sponge MPs in various natural water bodies (rivers, lakes, wetlands and seawater), simulated acidic, saline and alkaline water. The findings of this study indicate that the leaching of additives increases

Table 2
Release of additives and DOM during MPs degradation.

MPs	Additives/DOM	Time (d)	Release concentration	Reference
PE/PP PS PVC	(2,4-di-tert-butylphenyl) phosphite DEHP Diocetylterephthalate (DOTP) di(2-ethylhexyl) terephthalate	14	0.12–2.11 mg C L ⁻¹ in the dark 1.28–7.34 mg C L ⁻¹ under light conditions	Lee et al. (2023)
PVC PS	DEHP BPA	24	76.7 ± 0.6 mg C L ⁻¹ 11.2 ± 0.1 mg C L ⁻¹	Lee et al. (2020)
PE	TiO ₂	0.5	139.0 ± 1.4 mg L ⁻¹	Luo et al. (2020b)
PE PP Expanded PS (EPS)	DOM	54	EPS released the highest amount of DOC (68.2 ± 0.9 mg C L ⁻¹), followed by PP (39.1 ± 0.3 mg C L ⁻¹) and PE (1.1 ± 0.2 mg C L ⁻¹).	Zhu et al. (2020b)
PE PP EPS	Alicyclic compounds Aromatic compounds	54	–	Stubbins et al. (2023)
PE	Cr Pb	4	Under acidic condition (pH = 3.0), leaching of Cr is 18.3 ± 1.2 µg g ⁻¹ and the leaching of Pb is 81.4 ± 4.3 µg g ⁻¹	Luo et al. (2020a)
PS	Benzene Toluene Phenol	30	66 µg g ⁻¹	Wu et al. (2022)
PE PVC	di-isobutyl phthalate (DiBP) di-n-butyl phthalate (DnBP) dimethyl phthalate (DMP) diethyl phthalate (DEP)	84	PE bag released DiBP and DNBP, highest values 83.4 ± 12.5 and 120.1 ± 18.0 ng g ⁻¹ , respectively. PVC cables released DiBP and DNBP, with mass fractions 9.5 ± 1.4 and 68.9 ± 10.3 ng g ⁻¹ , respectively.	Paluselli et al. (2018)

with higher solution pH and longer leaching durations. This pH-dependent release of additives from degraded MPs demonstrates that the leaching process is mainly affected by plastic surface dissolution and protonation/deprotonation (Luo et al., 2019). Studies found that the salting out coefficient of di(2-ethylhexyl)phthalate (DEHP) increased with the increase of solution salinity. As salinity increases, the distribution coefficient $K_{\text{PVC/W}}$ of DEHP between PVC and water increases. The increase in flow rate result in higher rates of leaching. Compared with salinity, the impact of turbulence on leaching is more obvious (Henkel et al., 2023). The diffusion coefficients of UV stabilizers in LDPE are exceedingly low, whereas their partition coefficients between LDPE and water ($K_{\text{LDPE/W}}$) are very high. This indicates that UV stabilizers have limited mobility in aqueous environments. However, environmental factors play a more significant role in determining the actual leaching behavior than diffusion processes alone. The leaching kinetics of plastic additives may be altered by a variety of environmental factors, such as sunlight, mechanical forces, temperature, microbial invasion, etc. These factors tend to cause surface modification of MPs, thereby influencing the interactions between additives and polymers (Do et al., 2023).

Many additives and catalysts used in plastics contain toxic (and metalloid) compounds. Normally, metal additives do not chemically bind to polymers, allowing them to leach into the environment. In the aqueous phase, the release of these toxic metals will also negatively impact the environment (Turner and Filella, 2021). Acid washing of PVC has been shown to release heavy metals. After 24 h of PVC soaking, the content of Pb in the solution was $84.3 \pm 8.7 \mu\text{g/L}$ (Boyle et al., 2020). Researchers have assessed the release of metal contaminants from recycled plastic bottles following heating and microwave treatments (Cheng et al., 2010). MPs degradation can increase the release of pigments. As the degradation time increased, there was an observed rise in the release of pigment red in the simulated solution. Pigment red could bind to proteins and interact with them, causing fluorescence quenching of enzymes (Luo et al., 2020d). In fact, metals are released much faster in the plastics with size of hundreds of microns to millimeters, and the diffusion rate tends to stabilize over time (Smith and Turner, 2020). The additives released by MPs can also enter organisms through the food chain and cause harm to them. The effects of the leachate of seven MPs on barnacle larvae have been studied. There are plasticizers, stabilizers and other additives in the leachate. It was found that leachate from all seven types of MPs increased the mortality rate of barnacle larvae (Li et al., 2016).

In order to reduce the release of additives during the degradation process of MPs, the use of toxic additives should be avoided. It is advisable to seek out more environmentally friendly and biodegradable alternatives. Researchers may develop plastic products that are easier to degrade, thereby reducing their impact on the environment. To ensure compliance with environmental standards, it is necessary to reinforce the supervision of the production, use and recycling of plastics and additives. Understanding the release behavior and potential effect of additives during the degradation of MPs can provide scientific basis for making corresponding management and countermeasures.

5.2. Release of DOM

With the degradation of MPs, more DOM is released into the aquatic environment (Kwon et al., 2015). More than 100 types of DOM are released during MPs degradation. In the UV-based AOPs, the DOM released by MPs is mostly protein/phenol-like substances (Du et al., 2023). The presence of DOM enhances the photosensitivity of the MPs surface and facilitates the generation of free radicals, thereby accelerating the photochemical degradation process. It is found that MPs exposed directly to UV light release DOM with the higher fluorescence intensity than those in the dark. The generation of ROS during the photodegradation of MPs also promotes the release of DOM (Xu et al., 2022). Studies have shown that light-generated DOM contains oxygen atoms, indicating the formation of soluble oxygen-containing organic matter during photodegradation of plastics. The H/C ratio of PP and PE plastics was 2, and the average H/C ratio of DOM products was 1.7 ± 0.1 to 1.8 ± 0.1 . The H/C ratio of expanded polystyrene (EPS) was 1, and the average H/C ratio of DOM products was 0.9 ± 0.2 . This indicates that the stoichiometry of DOM generated from the photodegradation of MPs is related to the stoichiometry of polymers (Stubbins et al., 2023). Moreover, the release of DOM during MPs degradation differs among various polymer types. For instance, the photodegradation of PE and PP typically results in the generation of low molecular weight alkanes and alkenes due to the breakdown of long polymer chains (Bacha et al., 2021; Ranby, 1989). These smaller hydrocarbons, either as monomers or as oligomers, constitute a component of DOM in aquatic environments. In the case of PET, hydrolysis and oxidation release terephthalic acid, glycol monomers and oligomers as intermediates (Day and Wiles, 1972). For PS, oxidative photodegradation of the aromatic rings leads to the release of styrene monomers, as well as phenolic and carbonyl-containing derivatives (Xiao et al., 2014). PVC degradation primarily involves the dechlorination of the polymer, resulting in the release of chloroalkanes and oxygenated by-products (Wang et al.,

2020a). These monomers, oligomers and oxidation products not only alter the physicochemical properties of MPs, but also contribute to the complexity and reactivity of DOM in aquatic systems. Therefore, it is essential to consider the specific types of monomers and degradation intermediates released from different MPs in order to comprehensively evaluate the ecological impacts of MPs-derived DOM.

Leached DOM mainly consists of low molecular weight fractions, which contribute significantly to the formation of disinfection by-products (DBPs). DBPs are harmful by-products produced during water disinfection processes (Chen et al., 2023). The formation of DBPs is related to the composition of MPs-derived DOM (MPs-DOM) under oxidative conditions. MPs-DOM may be a precursor of DBPs according to previous reports (Wang et al., 2024). In addition, DOM can significantly inhibit the photodegradation of some organic pollutants, such as sulfamethoxazole, by inner filter effect and the complexation effect (Chen et al., 2022a). DOM released by MPs during the degradation process is toxic to the ecological environment. The aging test of PU-MPs showed that the concentration of PU-DOM increased after 14 days aging in water. The average survival rate of human colon adenocarcinoma cells decreased to 91 % upon exposure to PU-DOM (Xu et al., 2022). Overall, the leaching behavior of DOM during the degradation process of MPs involves a complex interplay of mechanisms including physical adsorption, chemical reaction, and biological degradation. An in-depth understanding of this process holds significant importance in assessing the environmental behavior and ecological risks of MPs and developing effective degradation technologies of MPs.

6. Conclusions and perspectives

The quantity of MPs in the environment is rapidly increasing, which has heightened concern over MPs pollution. Understanding the process of MPs degradation is highly significant for comprehensively assessing the ecological risks associated with them. This review summarizes in detail the various methods currently used to degrade MPs, including physical wear, electrochemical degradation, AOPs, photodegradation, photocatalytic degradation and biodegradation. In addition, DOM, some reducing agents and sulfides can also be used to degrade MPs in specific environment. The role of ROS in the chemical degradation of MPs and enzymes in biodegradation are revealed. Different types of MPs require different degrading enzymes and have different degradation pathways. This article details the specific degradation processes of different MPs. There will also be the leaching and diffusion behavior of intermediates such as additives and DOM during the degradation process of MPs, and their release process and toxic effects require further study. The intrinsic properties of MPs and external environmental conditions will affect the degradation behavior of MPs. This review analyzes the influence of the intrinsic characteristics of MPs and environmental factors on degradation. Factors influencing MPs degradation interact synergistically rather than independently, highlighting the need for comprehensive research into their interaction mechanisms.

On the basis of the existing research, future work is prospected and the following points are put forward to understand the environmental behavior of MPs and reduce the ecological risk.

- (1) At present, research on the degradation of MPs is still in its infancy. Many researchers indicate that their degradation efficiency is low and the degradation time is long. Therefore, it is crucial to conduct further research to ascertain the degradation mechanism and regulate the breakage of chemical bonds. The degradation of MPs requires high activity and high specificity catalyst. Alternatively, some biotechnology for the complete degradation of MPs could be developed. For example, modifying the enzyme gene sequence can enhance its selectivity and improve the degradation efficiency of MPs.
- (2) Many studies focus on specific types of plastics during degradation, which differs from real-world conditions where various

plastics coexist in water. We should develop an effective solution to address the degradation needs of a wide range of plastics, particularly through chemical processes, could achieve this purpose.

- (3) The incomplete degradation of MPs has raised widespread concerns about the generation of toxic by-products. The degradation of incompletely degraded MPs may release harmful chemicals that can cause damage to water bodies and their biological communities, thereby affecting balance of nature and negatively affecting human health. It is necessary to reveal the degradation pathways of MPs, identify the potential degradation products and assess their toxicity.
- (4) Currently, almost none of the MPs degradation methods are cost-effective. To overcome the shortcomings of cost and benefit, a new technology can also be developed to produce valuable products in the degradation process.

CRedit authorship contribution statement

Hongwei Luo: Writing – original draft, Resources, Methodology, Investigation. **Zhen Li:** Writing – original draft, Methodology, Investigation. **Yifeng Zeng:** Methodology, Investigation. **Dongqin He:** Resources, Methodology, Investigation. **Jianqiang Sun:** Writing – review & editing, Supervision. **Anping Zhang:** Writing – review & editing, Methodology, Funding acquisition. **Xiangliang Pan:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the National Key R&D Program of China (2023YFC3706604), National Natural Science Foundation of China (21806140, 51908499 and 42277407), China Postdoctoral Science Foundation (2019M662105 and 2021T140610), and Natural Science Foundation of Zhejiang Province (LY20B070010) for supporting this work.

Data availability

Data will be made available on request.

References

- Abdelfattah, A., Dar, M.A., Ramadan, H., El-Aswar, E.I., Eltwab, R., Abdelkarim, E., Elsamahy, T., Kornaros, M., Ali, S.S., Cheng, L., 2022. Handbook of research on algae as a sustainable solution for food, energy, and the environment. IGI Global 501–523.
- Aghilinasrollahabadi, K., Salehi, M., Fujiwara, T., 2021. Investigate the influence of microplastics weathering on their heavy metals uptake in stormwater. *J. Hazard. Mater.* 408, 124439.
- Albertsson, A.-C., Karlsson, S., 1993. Aspects of biodeterioration of inert and degradable polymers. *Int. Biodeterior. Biodegrad.* 31 (3), 161–170.
- Ali, S.S., Elsamahy, T., Al-Tohamy, R., Zhu, D., Mahmoud, Y.A.-G., Koutra, E., Metwally, M.A., Kornaros, M., Sun, J., 2021. Plastic wastes biodegradation: mechanisms, challenges and future prospects. *Sci. Total Environ.* 780, 146590.
- Alimi, O.S., Farner Budarz, J., Hernandez, L.M., Tufenkji, N., 2018. Microplastics and nanoplastics in aquatic environments: aggregation, deposition, and enhanced contaminant transport. *Environ. Sci. Technol.* 52 (4), 1704–1724.
- Allen, S., Materić, D., Allen, D., MacDonald, A., Holzinger, R., Roux, G.L., Phoenix, V.R., 2022. An early comparison of nano to microplastic mass in a remote catchment's atmospheric deposition. *J. Hazard. Mater. Adv.* 7, 100104.
- Anand, U., Dey, S., Bontempi, E., Ducoli, S., Vethaak, A.D., Dey, A., Federici, S., 2023. Biotechnological methods to remove microplastics: a review. *Environ. Chem. Lett.* 21 (3), 1787–1810.
- Andrady, A.L., 2017. The plastic in microplastics: a review. *Mar. Pollut. Bull.* 119 (1), 12–22.
- Andrady, A.L., 2022. Weathering and fragmentation of plastic debris in the ocean environment. *Mar. Pollut. Bull.* 180, 113761.
- Ariza-Tarazona, M.C., Villarreal-Chiu, J.F., Barbieri, V., Siligardi, C., Cedillo-González, E. I., 2019. New strategy for microplastic degradation: green photocatalysis using a protein-based porous N-TiO₂ semiconductor. *Ceram. Int.* 45 (7, Part B), 9618–9624.
- Ariza-Tarazona, M.C., Villarreal-Chiu, J.F., Hernández-López, J.M., De la Rosa, J.R., Barbieri, V., Siligardi, C., Cedillo-González, E.I., 2020. Microplastic pollution reduction by a carbon and nitrogen-doped TiO₂: effect of pH and temperature in the photocatalytic degradation process. *J. Hazard. Mater.* 395, 122632.
- Artham, T., Doble, M., 2008. Biodegradation of aliphatic and aromatic polycarbonates. *Macromol. Biosci.* 8 (1), 14–24.
- Bacha, A.-U.-R., Nabi, I., Zhang, L., 2021. Mechanisms and the engineering approaches for the degradation of microplastics. *ACS ES&T Eng.* 1 (11), 1481–1501.
- Bai, B., Jin, H., Fan, C., Cao, C., Wei, W., Cao, W., 2019. Experimental investigation on liquefaction of plastic waste to oil in supercritical water. *Waste Manage. (Tucson, Ariz.)* 89, 247–253.
- Barlow, D.E., Biffinger, J.C., Estrella, L., Lu, Q., Hung, C.-S., Nadeau, L.J., Crouch, A.L., Russell, Jr J.N., Crookes-Goodson, W.J., 2020. Edge-localized biodeterioration and secondary microplastic formation by *Papiliotrema laurentii* unsaturated biofilm cells on polyurethane films. *Langmuir* 36 (6), 1596–1607.
- Beltrán, F.J., Álvarez, P.M., Gimeno, O., 2019. Graphene-based catalysts for ozone processes to decontaminate water. *Molecules* 24 (19), 3438.
- Biale, G., La Nasa, J., Mattonai, M., Corti, A., Castelvetro, V., Modugno, F., 2022. Seeping plastics: potentially harmful molecular fragments leaching out from microplastics during accelerated ageing in seawater. *Water Res.* 219, 118521.
- Bikiaris, D.N., Karayannidis, G.P., 1999. Effect of carboxylic end groups on thermooxidative stability of PET and PBT. *Polym. Degrad. Stabil.* 63 (2), 213–218.
- Bockhorn, H., Hornung, A., Hornung, U., Schwallier, D., 1999. Kinetic study on the thermal degradation of polypropylene and polyethylene. *J. Anal. Appl. Pyrolysis* 48 (2), 93–109.
- Boyle, D., Catarino, A.I., Clark, N.J., Henry, T.B., 2020. Polyvinyl chloride (PVC) plastic fragments release Pb additives that are bioavailable in zebrafish. *Environ. Pollut.* 263, 114422.
- Calvert, P.D., Chalivendra, V., Wanasekara, N., 2011. Sub-micron scale mechanical properties of polypropylene fibers exposed to ultraviolet and thermal degradation. *Polym. Degrad. Stabil.* 96, 432–437.
- Chen, M., Xu, J., Tang, R., Yuan, S., Min, Y., Xu, Q., Shi, P., 2022a. Roles of microplastic-derived dissolved organic matter on the photodegradation of organic micropollutants. *J. Hazard. Mater.* 440, 129784.
- Chen, Y., Qian, Y., An, D., Liu, F., Vollertsen, J., Gan, J., 2023. Potential disinfection byproducts-related risks to drinking water? Molecular insights into the dissolved organic matter from photodegradation of polyethylene microplastics. *ACS ES&T Water* 4 (1), 217–226.
- Chen, Z., Liu, Y., Wei, W., Ni, B.-J., 2019. Recent advances in electrocatalysts for halogenated organic pollutant degradation. *Environ. Sci. Nano* 6 (8), 2332–2366.
- Chen, Z., Wei, W., Liu, X., Ni, B.-J., 2022b. Emerging electrochemical techniques for identifying and removing micro/nanoplastics in urban waters. *Water Res.* 221, 118846.
- Cheng, X., Shi, H., Adams, C.D., Ma, Y., 2010. Assessment of metal contaminations leaching out from recycling plastic bottles upon treatments. *Environ. Sci. Pollut. Res.* 17 (7), 1323–1330.
- Christian, D., Michel, B., 1982. Degradation of poly(vinyl chloride) by U.V. radiation—I. Kinetics and quantum yields. *Eur. Polym. J.* 18 (12), 1085–1091.
- Costa, J.P.d., Avellan, A., Mouneyrac, C., Duarte, A., Rocha-Santos, T., 2023. Plastic additives and microplastics as emerging contaminants: mechanisms and analytical assessment. *TrAC, Trends Anal. Chem.* 158, 116898.
- Dai, J., Liu, P., Wang, C., Li, H., Qiang, H., Yang, Z., Guo, X., Gao, S., 2023. Which factors mainly drive the photoaging of microplastics in freshwater? *Sci. Total Environ.* 858, 159845.
- Danso, D., Chow, J., Streit, W.R., 2019. Plastics: environmental and biotechnological perspectives on microbial degradation. *Appl. Environ. Microbiol.* 85 (19), 01095–01019.
- Das, M.P., Kumar, S., Das, J., 2018. Fungal-mediated deterioration and biodegradation study of low-density polyethylene (LDPE) isolated from municipal dump yard in Chennai, India. *Energy, Ecol. Environ.* 3 (4), 229–236.
- Datta, C., Varun Kondra, T., Miller, M., Streltsov, A., 2023. Catalysis of entanglement and other quantum resources. *Rep. Prog. Phys.* 86 (11), 116002.
- Day, M., Wiles, D., 1972. Photochemical degradation of poly (ethylene terephthalate). III. Determination of decomposition products and reaction mechanism. *J. Appl. Polym. Sci.* 16 (1), 203–215.
- Decker, C., Balandier, M., 1981. Laser-induced degradation of polyvinyl chloride II: oxygen bleaching of polyenes. *J. Photochem.* 15 (3), 221–228.
- Deng, Y., Zhu, K., Sun, Y., Xie, L., Jiang, W., Jiang, Y., Guo, X., Jia, H., 2024. Aging kinetics and mechanisms of polystyrene microplastic in water under sunlight irradiation: effects of inorganic cations. *Gondwana Res.* 129, 193–200.
- Ding, L., Luo, Y., Yu, X., Ouyang, Z., Liu, P., Guo, X., 2022a. Insight into interactions of polystyrene microplastics with different types and compositions of dissolved organic matter. *Sci. Total Environ.* 824, 153883.
- Ding, T., Wei, L., Hou, Z., Li, J., Zhang, C., Lin, D., 2022b. Microplastics altered contaminant behavior and toxicity in natural waters. *J. Hazard. Mater.* 425, 127908.
- Djakhane, K., Dehbi, A., Mourad, A.I., Zaoui, A., Picuno, P., 2016. The effect of sand wind, temperature and exposure time on tri-layer polyethylene film used as greenhouse roof. *Plast. Rubber Compos.* 45 (8), 346–351.
- Do, A.T.N., Ha, Y., Kwon, J.-H., 2023. Effect of mass fraction on leaching kinetics of hydrophobic ultraviolet stabilizers in low-density polyethylene. *Environ. Sci. Technol.* 57, 21428–21437.

- Dos Santos, N.d.O., Busquets, R., Campos, L.C., 2023. Insights into the removal of microplastics and microfibrils by advanced oxidation processes. *Sci. Total Environ.* 861, 160665.
- Du, H., Xie, Y., Wang, J., 2021. Microplastic degradation methods and corresponding degradation mechanism: research status and future perspectives. *J. Hazard. Mater.* 418, 126377.
- Du, Z., Li, G., Ding, S., Song, W., Zhang, M., Jia, R., Chu, W., 2023. Effects of UV-based oxidation processes on the degradation of microplastic: fragmentation, organic matter release, toxicity and disinfection byproduct formation. *Water Res.* 237, 119983.
- Duan, J., Li, Y., Gao, J., Cao, R., Shang, E., Zhang, W., 2022. ROS-mediated photoaging pathways of nano-and micro-plastic particles under UV irradiation. *Water Res.* 216, 118320.
- Duan, L., Qin, Y., Meng, X., Liu, Y., Zhang, T., Chen, W., 2023. Sulfide- and UV-induced aging differentially affect contaminant-binding properties of microplastics derived from commercial plastic products. *Sci. Total Environ.* 869, 161800.
- Dümichen, E., Eisentraut, P., Bannick, C.G., Barthel, A.-K., Senz, R., Braun, U., 2017. Fast identification of microplastics in complex environmental samples by a thermal degradation method. *Chemosphere* 174, 572–584.
- Edge, M., Hayes, M., Mohammadian, M., Allen, N., Jewitt, T., Brems, K., Jones, K., 1991. Aspects of poly (ethylene terephthalate) degradation for archival life and environmental degradation. *Polym. Degrad. Stabil.* 32 (2), 131–153.
- El-Dash, H.A., Yousef, N.E., Aboelazm, A.A., Awan, Z.A., Yahya, G., El-Ganiny, A.M., 2023. Optimizing eco-friendly degradation of polyvinyl chloride (PVC) plastic using environmental strains of *Malassezia* species and *Aspergillus fumigatus*. *Int. J. Mol. Sci.* 24 (20), 15452.
- Elizalde-Velázquez, G.A., Gómez-Oliván, L.M., 2021. Microplastics in aquatic environments: a review on occurrence, distribution, toxic effects, and implications for human health. *Sci. Total Environ.* 780, 146551.
- Enfrin, M., Hachemi, C., Hodgson, P.D., Jegatheesan, V., Vrouwenvelder, J., Callahan, D. L., Lee, J., Dumée, L.F., 2021. Nano/micro plastics—challenges on quantification and remediation: a review. *J. Water Process Eng.* 42, 102128.
- Fa, W., Wang, J., Ge, S., Chao, C., 2020. Performance of photo-degradation and thermo-degradation of polyethylene with photo-catalysts and thermo-oxidant additives. *Polym. Bull.* 77, 1417–1432.
- García-Montelongo, X., Martínez-De La Cruz, A., Vázquez-Rodríguez, S., Torres-Martínez, L.M., 2014. Photo-oxidative degradation of TiO₂/polypropylene films. *Mater. Res. Bull.* 51, 56–62.
- Gautam, R., Bassi, A., Yanful, E., 2007. A review of biodegradation of synthetic plastic and foams. *Appl. Biochem. Biotechnol.* 141, 85–108.
- Ge, J., Wang, M., Liu, P., Zhang, Z., Peng, J., Guo, X., 2023. A systematic review on the aging of microplastics and the effects of typical factors in various environmental media. *TrAC, Trends Anal. Chem.* 162, 117025.
- Ge, J., Zhang, Z., Ouyang, Z., Shang, M., Liu, P., Li, H., Guo, X., 2022. Photocatalytic degradation of (micro)plastics using TiO₂-based and other catalysts: properties, influencing factor, and mechanism. *Environ. Res.* 209, 112729.
- Gewert, B., Plassmann, M.M., MacLeod, M., 2015. Pathways for degradation of plastic polymers floating in the marine environment. *Environ. Sci.: Process. Impacts* 17 (9), 1513–1521.
- Grigorakis, S., Mason, S.A., Drouillard, K.G., 2017. Determination of the gut retention of plastic microbeads and microfibrils in goldfish (*Carassius auratus*). *Chemosphere* 169, 233–238.
- Gunaalan, K., Fabbri, E., Capolupo, M., 2020. The hidden threat of plastic leachates: a critical review on their impacts on aquatic organisms. *Water Res.* 184, 116170.
- Guo, X., Wang, J., 2019. Sorption of antibiotics onto aged microplastics in freshwater and seawater. *Mar. Pollut. Bull.* 149, 110511.
- Hahladakis, J.N., Velis, C.A., Weber, R., Iacovidou, E., Purnell, P., 2018. An overview of chemical additives present in plastics: migration, release, fate and environmental impact during their use, disposal and recycling. *J. Hazard. Mater.* 344, 179–199.
- Harris, P.T., 2020. The fate of microplastic in marine sedimentary environments: a review and synthesis. *Mar. Pollut. Bull.* 158, 111398.
- He, J., Ma, W., Han, L., Chen, L., Xu, E.G., Xing, B., Yang, Z., 2023a. Unraveling the role of natural and pyrogenic dissolved organic matter in photodegradation of biodegradable microplastics in freshwater. *Carbon Res.* 2 (1), 18.
- He, W., Liu, S., Zhang, W., Yi, K., Zhang, C., Pang, H., Huang, D., Huang, J., Li, X., 2023b. Recent advances on microplastic aging: identification, mechanism, influence factors, and additives release. *Sci. Total Environ.* 889, 164035.
- Hebner, T.S., Maurer-Jones, M.A., 2020. Characterizing microplastic size and morphology of photodegraded polymers placed in simulated moving water conditions. *Environ. Sci.: Process. Impacts* 22 (2), 398–407.
- Henkel, C., Lamprecht, J., Hüffer, T., Hofmann, T., 2023. Environmental factors strongly influence the leaching of di(2-ethylhexyl) phthalate from polyvinyl chloride microplastics. *Water Res.* 242, 120235.
- Hermabessiere, L., Dehaut, A., Paul-Pont, I., Lacroix, C., Jezequel, R., Soudant, P., Duflos, G., 2017. Occurrence and effects of plastic additives on marine environments and organisms: a review. *Chemosphere* 182, 781–793.
- Ho, B.T., Roberts, T.K., Lucas, S., 2018. An overview on biodegradation of polystyrene and modified polystyrene: the microbial approach. *Crit. Rev. Biotechnol.* 38 (2), 308–320.
- Horne, F.J., Liggat, J.J., MacDonald, W.A., Sankey, S.W., 2020. Photo-oxidation of poly (ethylene terephthalate) films intended for photovoltaic backsheets. *J. Appl. Polym. Sci.* 137 (17), 48623.
- Hu, K., Zhou, P., Yang, Y., Hall, T., Nie, G., Yao, Y., Duan, X., Wang, S., 2022a. Degradation of microplastics by a thermal fenton reaction. *ACS ES&T Eng.* 2 (1), 110–120.
- Hu, T., Tang, L., Feng, H., Zhang, J., Li, X., Zuo, Y., Lu, Z., Tang, W., 2022b. Metal-organic frameworks (MOFs) and their derivatives as emerging catalysts for electro-Fenton process in water purification. *Coord. Chem. Rev.* 451, 214277.
- Jean-Luc, G., Jacques, L., 1991. Photothermal and thermal oxidations of rigid, plasticized and pigmented poly(vinyl chloride). *Polym. Degrad. Stabil.* 34 (1), 135–167.
- Jiang, R., Lu, G., Yan, Z., Liu, J., Wu, D., Wang, Y., 2021. Microplastic degradation by hydroxy-rich bismuth oxychloride. *J. Hazard. Mater.* 405, 124247.
- Jin, Y., Xia, J., Pan, Z., Yang, J., Wang, W., Fu, Z., 2018. Polystyrene microplastics induce microbiota dysbiosis and inflammation in the gut of adult zebrafish. *Environ. Pollut.* 235, 322–329.
- Kanaga Karupiah, K.S., Bruck, A.L., Sundararajan, S., Wang, J., Lin, Z., Xu, Z.-H., Li, X., 2008. Friction and wear behavior of ultra-high molecular weight polyethylene as a function of polymer crystallinity. *Acta Biomater.* 4 (5), 1401–1410.
- Kang, J., Zhou, L., Duan, X., Sun, H., Ao, Z., Wang, S., 2019. Degradation of cosmetic microplastics via functionalized carbon nanosprings. *Matter* 1 (3), 745–758.
- Kang, M.-G., Kwak, M.-J., Kim, Y.J., 2023. Polystyrene microplastics biodegradation by gut bacterial *Enterobacter hormaechei* from mealworms under anaerobic conditions: anaerobic oxidation and depolymerization. *J. Hazard. Mater.* 459, 132045.
- Karalija, E., Carbo, M., Coppi, A., Colzi, I., Dainelli, M., Gašparović, M., Grebenc, T., Gonnelli, C., Papadakis, V., Pilić, S., Sibanc, N., Valledor, L., Poma, A., Martinelli, F., 2022. Interplay of plastic pollution with algae and plants: hidden danger or a blessing? *J. Hazard. Mater.* 438, 129450.
- Khandare, S.D., Chaudhary, D.R., Jha, B., 2021. Bioremediation of polyvinyl chloride (PVC) films by marine bacteria. *ar. Pollut. Bull.* 169, 112566.
- Khatoun, N., Jamal, A., Ali, M.I., 2019. Lignin peroxidase isoenzyme: a novel approach to biodegrade the toxic synthetic polymer waste. *Environ. Technol.* 40 (11), 1366–1375.
- Kiendreebo, M., Karimi Estahbanati, M.R., Ouara, Y., Drogui, P., Tyagi, R.D., 2022. Electrochemical degradation of nanoplastics in water: analysis of the role of reactive oxygen species. *Sci. Total Environ.* 808, 151897.
- Kim, H.R., Lee, H.M., Yu, H.C., Jeon, E., Lee, S., Li, J., Kim, D.-H., 2020. Biodegradation of polystyrene by *Pseudomonas* sp. isolated from the gut of superworms (larvae of *Zophobas atratus*). *Environ. Sci. Technol.* 54 (11), 6987–6996.
- Kim, M.S., Kim, H.-H., Lee, K.-M., Lee, H.-J., Lee, C., 2017. Oxidation of microcystin-LR by ferrous-tetrapolyphosphate in the presence of oxygen and hydrogen peroxide. *Water Res.* 114, 277–285.
- Kong, F., Xu, X., Xue, Y., Gao, Y., Zhang, L., Wang, L., Jiang, S., Zhang, Q., 2021. Investigation of the adsorption of sulfamethoxazole by degradable microplastics artificially aged by chemical oxidation. *Arch. Environ. Contam. Toxicol.* 81 (1), 155–165.
- Kowalski, N., Reichardt, A.M., Wanek, J.J., 2016. Sinking rates of microplastics and potential implications of their alteration by physical, biological, and chemical factors. *Mar. Pollut. Bull.* 109 (1), 310–319.
- Krupp, L.R., Jewell, W.J., 1992. Biodegradability of modified plastic films in controlled biological environments. *Environ. Sci. Technol.* 26 (1), 193–198.
- Kwon, B.G., Koizumi, K., Chung, S.-Y., Kodera, Y., Kim, J.-O., Saido, K., 2015. Global styrene oligomers monitoring as new chemical contamination from polystyrene plastic marine pollution. *J. Hazard. Mater.* 300, 359–367.
- Lambert, S., Sinclair, C., Boxall, A., 2013. Occurrence, degradation, and effect of polymer-based materials in the environment. *Rev. Environ. Contam. Toxicol.* 227, 1–53.
- Lameh, F., Baseer, A.Q., Ashiru, A.G., 2022. Retracted: comparative molecular docking and molecular-dynamic simulation of wild-type and mutant carboxylesterase with BTA-hydrolase for enhanced binding to plastic. *Eng. Life Sci.* 22 (1), 13–29.
- Lee, Y.K., He, W., Guo, H., Karanfil, T., Hur, J., 2023. Effects of organic additives on spectroscopic and molecular-level features of photo-induced dissolved organic matter from microplastics. *Water Res.* 242, 120272.
- Lee, Y.K., Murphy, K.R., Hur, J., 2020. Fluorescence signatures of dissolved organic matter leached from microplastics: polymers and additives. *Environ. Sci. Technol.* 54 (19), 11905–11914.
- Li, C., Jiang, B., Guo, J., Sun, C., Shi, C., Huang, S., Liu, W., Wu, C., Zhang, Y., 2022. Aging process of microplastics in the aquatic environments: aging pathway, characteristic change, compound effect, and environmentally persistent free radicals formation. *Water* 14 (21), 3515.
- Li, H.-X., Getzinger, G.J., Ferguson, P.L., Orihuela, B., Zhu, M., Rittschof, D., 2016. Effects of toxic leachate from commercial plastics on larval survival and settlement of the barnacle amphibalanus amphitrite. *Environ. Sci. Technol.* 50 (2), 924–931.
- Li, R., Liu, Y., Sheng, Y., Xiang, Q., Zhou, Y., Cizdziel, J.V., 2020. Effect of prothioconazole on the degradation of microplastics derived from mulching plastic film: apparent change and interaction with heavy metals in soil. *Environ. Pollut.* 260, 113988.
- Liu, G., Zhu, Z., Yang, Y., Sun, Y., Yu, F., Ma, J., 2019a. Sorption behavior and mechanism of hydrophilic organic chemicals to virgin and aged microplastics in freshwater and seawater. *Environ. Pollut.* 246, 26–33.
- Liu, L., Xu, M., Ye, Y., Zhang, B., 2022a. On the degradation of (micro)plastics: degradation methods, influencing factors, environmental impacts. *Sci. Total Environ.* 806, 151312.
- Liu, P., Li, H., Wu, J., Wu, X., Shi, Y., Yang, Z., Huang, K., Guo, X., Gao, S., 2022b. Polystyrene microplastics accelerated photodegradation of co-existed polypropylene via photosensitization of polymer itself and released organic compounds. *Water Res.* 214, 118209.
- Liu, P., Qian, L., Wang, H., Zhan, X., Lu, K., Gu, C., Gao, S., 2019b. New insights into the aging behavior of microplastics accelerated by advanced oxidation processes. *Environ. Sci. Technol.* 53 (7), 3579–3588.

- Liu, P., Shi, Y., Wu, X., Wang, H., Huang, H., Guo, X., Gao, S., 2021a. Review of the artificially-accelerated aging technology and ecological risk of microplastics. *Sci. Total Environ.* 768, 144969.
- Liu, P., Zhan, X., Wu, X., Li, J., Wang, H., Gao, S., 2020. Effect of weathering on environmental behavior of microplastics: properties, sorption and potential risks. *Chemosphere* 242, 125193.
- Liu, Y., Zhao, Y., Wang, J., 2021b. Fenton/Fenton-like processes with in-situ production of hydrogen peroxide/hydroxyl radical for degradation of emerging contaminants: advances and prospects. *J. Hazard. Mater.* 404, 124191.
- Lombardo, S.M., Vindedahl, A.M., Arnold, W.A., 2020. Determination of hydroxyl radical production from sulfide oxidation relevant to sulfidic porewaters. *ACS Earth Space Chem.* 4 (2), 261–271.
- Lomonaco, T., Manco, E., Corti, A., La Nasa, J., Ghimenti, S., Biagini, D., Di Francesco, F., Modugno, F., Ceccarini, A., Fuoco, R., Castelvetro, V., 2020. Release of harmful volatile organic compounds (VOCs) from photo-degraded plastic debris: a neglected source of environmental pollution. *J. Hazard. Mater.* 394, 122596.
- Lu, J., Hou, R., Wang, Y., Zhou, L., Yuan, Y., 2022. Surfactant-sodium dodecyl sulfate enhanced degradation of polystyrene microplastics with an energy-saving electrochemical advanced oxidation process (EAOP) strategy. *Water Res.* 226, 119277.
- Lu, Q., Zhou, Y., Sui, Q., Zhou, Y., 2023. Mechanism and characterization of microplastic aging process: a review. *Front. Environ. Sci. Eng.* 17 (8), 100.
- Lucas, N., Bienaime, C., Belloy, C., Queneudec, M., Silvestre, F., Nava-Saucedo, J.-E., 2008. Polymer biodegradation: mechanisms and estimation techniques—A review. *Chemosphere* 73 (4), 429–442.
- Luo, H., Li, Y., Zhao, Y., Xiang, Y., He, D., Pan, X., 2020a. Effects of accelerated aging on characteristics, leaching, and toxicity of commercial lead chromate pigmented microplastics. *Environ. Pollut.* 257, 113475.
- Luo, H., Liu, C., He, D., Sun, J., Li, J., Pan, X., 2022a. Effects of aging on environmental behavior of plastic additives: migration, leaching, and ecotoxicity. *Sci. Total Environ.* 849, 157951.
- Luo, H., Liu, C., He, D., Sun, J., Zhang, A., Li, J., Pan, X., 2022b. Interactions between polypropylene microplastics (PP-MPs) and humic acid influenced by aging of MPs. *Water Res.* 222, 118921.
- Luo, H., Liu, C., He, D., Xu, J., Sun, J., Li, J., Pan, X., 2022c. Environmental behaviors of microplastics in aquatic systems: a systematic review on degradation, adsorption, toxicity and biofilm under aging conditions. *J. Hazard. Mater.* 423, 126915.
- Luo, H., Tu, C., He, D., Zhang, A., Sun, J., Li, J., Xu, J., Pan, X., 2023. Interactions between microplastics and contaminants: a review focusing on the effect of aging process. *Sci. Total Environ.* 899, 165615.
- Luo, H., Tu, C., Liu, C., Zeng, Y., He, D., Zhang, A., Xu, J., Pan, X., 2024. Probing the molecular interaction between photoaged polystyrene microplastics and fulvic acid. *Sci. Total Environ.* 920, 170933.
- Luo, H., Xiang, Y., He, D., Li, Y., Zhao, Y., Wang, S., Pan, X., 2019. Leaching behavior of fluorescent additives from microplastics and the toxicity of leachate to *Chlorella vulgaris*. *Sci. Total Environ.* 678, 1–9.
- Luo, H., Xiang, Y., Li, Y., Zhao, Y., Pan, X., 2020b. Weathering alters surface characteristic of TiO₂-pigmented microplastics and particle size distribution of TiO₂ released into water. *Sci. Total Environ.* 729, 139083.
- Luo, H., Xiang, Y., Li, Y., Zhao, Y., Pan, X., 2021a. Photocatalytic aging process of Nano-TiO₂ coated polypropylene microplastics: combining atomic force microscopy and infrared spectroscopy (AFM-IR) for nanoscale chemical characterization. *J. Hazard. Mater.* 404, 124159.
- Luo, H., Xiang, Y., Zhao, Y., Li, Y., Pan, X., 2020c. Nanoscale infrared, thermal and mechanical properties of aged microplastics revealed by an atomic force microscopy coupled with infrared spectroscopy (AFM-IR) technique. *Sci. Total Environ.* 744, 140944.
- Luo, H., Zeng, Y., Zhao, Y., Xiang, Y., Li, Y., Pan, X., 2021b. Effects of advanced oxidation processes on leachates and properties of microplastics. *J. Hazard. Mater.* 413, 125342.
- Luo, H., Zhao, Y., Li, Y., Xiang, Y., He, D., Pan, X., 2020d. Aging of microplastics affects their surface properties, thermal decomposition, additives leaching and interactions in simulated fluids. *Sci. Total Environ.* 714, 136862.
- Lv, S., Li, Y., Zhao, S., Shao, Z., 2024. Biodegradation of Typical Plastics: from Microbial Diversity to Metabolic Mechanisms.
- Matjašić, T., Simčić, T., Medvešek, N., Bajt, O., Dreo, T., Mori, N., 2021. Critical evaluation of biodegradation studies on synthetic plastics through a systematic literature review. *Sci. Total Environ.* 752, 141959.
- Nabi, I., Bacha, A.-U.-R., Ahmad, F., Zhang, L., 2021. Application of titanium dioxide for the photocatalytic degradation of macro- and micro-plastics: a review. *J. Environ. Chem. Eng.* 9 (5), 105964.
- Nunes, R.W., Martin, J.R., Johnson, J.F., 1982. Influence of molecular weight and molecular weight distribution on mechanical properties of polymers. *Polym. Eng. Sci.* 22 (4), 205–228.
- Oehlmann, J., Schulte-Oehlmann, U., Kloas, W., Jagnytsch, O., Lutz, I., Kusk, K.O., Wollenberger, L., Santos, E.M., Paull, G.C., Van Look, K.J.W., Tyler, C.R., 2009. A critical analysis of the biological impacts of plasticizers on wildlife. *Philos. Trans. R. Soc. B* 364 (1526), 2047–2062.
- Othman, A.R., Hasan, H.A., Muhamad, M.H., Ismail, N.I., Abdullah, S.R.S., 2021. Microbial degradation of microplastics by enzymatic processes: a review. *Environ. Chem. Lett.* 19, 3057–3073.
- Ouyang, Z., Li, S., Xue, J., Liao, J., Xiao, C., Zhang, H., Li, X., Liu, P., Hu, S., Guo, X., Zhu, L., 2023. Dissolved organic matter derived from biodegradable microplastic promotes photo-aging of coexisting microplastics and alters microbial metabolism. *J. Hazard. Mater.* 445, 130564.
- Paluselli, A., Fauvel, V., Galgani, F., Sempéré, R., 2018. Phthalate release from plastic fragments and degradation in seawater. *Environ. Sci. Technol.* 53 (1), 166–175.
- Peng, X., Chen, M., Chen, S., Dasgupta, S., Xu, H., Ta, K., Du, M., Li, J., Guo, Z., Bai, S., 2018. Microplastics contaminate the deepest part of the world's ocean. *Geochem. Perspect. Lett.* 1–5.
- Qiu, X., Ma, S., Zhang, J., Fang, L., Guo, X., Zhu, L., 2022. Dissolved organic matter promotes the aging process of polystyrene microplastics under dark and ultraviolet light conditions: the crucial role of reactive oxygen species. *Environ. Sci. Technol.* 56 (14), 10149–10160.
- Ranby, B., 1989. Photodegradation and photo-oxidation of synthetic polymers. *J. Anal. Appl. Pyrolysis* 15, 237–247.
- Ravishanker, K., Ramesh, P.S., Sadhasivam, B., Raghavachari, D., 2018. Wear-induced mechanical degradation of plastics by low-energy wet-grinding. *Polym. Degrad. Stab.* 158, 212–219.
- Ren, Z., Gui, X., Wei, Y., Chen, X., Xu, X., Zhao, L., Qiu, H., Cao, X., 2021. Chemical and photo-initiated aging enhances transport risk of microplastics in saturated soils: key factors, mechanisms, and modeling. *Water Res.* 202, 117407.
- Romera-Castillo, C., Pinto, M., Langer, T.M., Álvarez-Salgado, X.A., Herndl, G.J., 2018. Dissolved organic carbon leaching from plastics stimulates microbial activity in the ocean. *Nat. Commun.* 9 (1), 1430.
- Ru, J., Huo, Y., Yang, Y., 2020. Microbial degradation and valorization of plastic wastes. *Front. Microbiol.* 11, 507487.
- Schmidt, J., Wei, R., Oeser, T., Dedavid e Silva, L.A., Breite, D., Schulze, A., Zimmermann, W., 2017. Degradation of polyester polyurethane by bacterial polyester hydrolases. *Polymers* 9 (2), 65.
- Shah, Z., Krumholz, L., Aktas, D.F., Hasan, F., Khattak, M., Shah, A.A., 2013. Degradation of polyester polyurethane by a newly isolated soil bacterium, *Bacillus subtilis* strain MZA-75. *Biodegradation* 24, 865–877.
- Shen, M., Song, B., Zhou, C., Hu, T., Zeng, G., Zhang, Y., 2022. Advanced oxidation processes for the elimination of microplastics from aqueous systems: assessment of efficiency, perspectives and limitations. *Sci. Total Environ.* 842, 156723.
- Shi, Y., Liu, P., Wu, X., Shi, H., Huang, H., Wang, H., Gao, S., 2021. Insight into chain scission and release profiles from photodegradation of polycarbonate microplastics. *Water Res.* 195, 116980.
- Sivan, A., 2011. New perspectives in plastic biodegradation. *Curr. Opin. Biotechnol.* 22 (3), 422–426.
- Smith, E.C., Turner, A., 2020. Mobilisation kinetics of Br, Cd, Cr, Hg, Pb and Sb in microplastics exposed to simulated, dietary-adapted digestive conditions of seabirds. *Sci. Total Environ.* 733, 138802.
- Stubbins, A., Zhu, L., Zhao, S., Spencer, R.G., Podgorski, D.C., 2023. Molecular signatures of dissolved organic matter generated from the photodissolution of microplastics in sunlight seawater. *Environ. Sci. Technol.* 57 (48), 20097–20106.
- Sun, Y., Yuan, J., Zhou, T., Zhao, Y., Yu, F., Ma, J., 2020. Laboratory simulation of microplastics weathering and its adsorption behaviors in an aqueous environment: a systematic review. *Environ. Pollut.* 265, 114864.
- Sutkar, P.R., Gadewar, R.D., Dhulap, V.P., 2023. Recent trends in degradation of microplastics in the environment: a state-of-the-art review. *J. Hazard. Mater. Adv.* 11, 100343.
- Tamnoui, E.B.M., Arfao, A.T., Nougang, M.E., Metsopkeng, C.S., Ewoti, O.V.N., Mounang, L.M., Nana, P.A., Takang-Etta, L.-R.A., Perrière, F., Sime-Ngando, T., 2021. Biodegradation of polyethylene by the bacterium *Pseudomonas aeruginosa* in acidic aquatic microcosm and effect of the environmental temperature. *Environ. Chall.* 3, 100056.
- Taniguchi, I., Yoshida, S., Hiraga, K., Miyamoto, K., Kimura, Y., Oda, K., 2019. Biodegradation of PET: current status and application aspects. *ACS Catal.* 9 (5), 4089–4105.
- Thakur, B., Singh, J., Singh, J., Angmo, D., Vig, A.P., 2023. Biodegradation of different types of microplastics: molecular mechanism and degradation efficiency. *Sci. Total Environ.* 877, 162912.
- Tiso, T., Narancic, T., Wei, R., Pollet, E., Beagan, N., Schröder, K., Honak, A., Jiang, M., Kenny, S.T., Wierckx, N., Perrin, R., Avérous, L., Zimmermann, W., O'Connor, K., Blank, L.M., 2021. Towards bio-upcycling of polyethylene terephthalate. *Metab. Eng.* 66, 167–178.
- Tofa, T.S., Kunjali, K.L., Paul, S., Dutta, J., 2019a. Visible light photocatalytic degradation of microplastic residues with zinc oxide nanorods. *Environ. Chem. Lett.* 17 (3), 1341–1346.
- Tofa, T.S., Ye, F., Kunjali, K.L., Dutta, J., 2019b. Enhanced visible light photodegradation of microplastic fragments with plasmonic platinum/zinc oxide nanorod photocatalysts. *Catalysts* 9 (10), 819.
- Turchi, C.S., Ollis, D.F., 1990. Photocatalytic degradation of organic water contaminants: mechanisms involving hydroxyl radical attack. *J. Catal.* 122 (1), 178–192.
- Turner, A., Filella, M., 2021. Hazardous metal additives in plastics and their environmental impacts. *Environ. Int.* 156, 106622.
- Uheida, A., Mejia, H.G., Abdel-Rehim, M., Hamd, W., Dutta, J., 2021. Visible light photocatalytic degradation of polypropylene microplastics in a continuous water flow system. *J. Hazard. Mater.* 406, 124299.
- Vital-Grappin, A.D., Ariza-Tarazona, M.C., Luna-Hernández, V.M., Villarreal-Chiu, J.F., Hernández-López, J.M., Siligardi, C., Cedillo-González, E.I., 2021. The role of the reactive species involved in the photocatalytic degradation of hdpe microplastics using C, N-TiO₂ powders. *Polymers* 13 (7), 999.
- Wang, C., Xian, Z., Jin, X., Liang, S., Chen, Z., Pan, B., Wu, B., Ok, Y.S., Gu, C., 2020a. Photo-aging of polyvinyl chloride microplastic in the presence of natural organic acids. *Water Res.* 183, 116082.
- Wang, Q., Chen, M., Min, Y., Shi, P., 2024. Aging of polystyrene microplastics by UV/Sodium percarbonate oxidation: organic release, mechanism, and disinfection by-product formation. *J. Hazard. Mater.* 464, 132934.

- Wang, X., Zheng, H., Zhao, J., Luo, X., Wang, Z., Xing, B., 2020b. Photodegradation elevated the toxicity of polystyrene microplastics to grouper (*Epinephelus moara*) through disrupting hepatic lipid homeostasis. *Environ. Sci. Technol.* 54 (10), 6202–6212.
- Wei, X.-F., Capezza, A.J., Cui, Y., Li, L., Hakonen, A., Liu, B., Hedenqvist, M.S., 2022. Millions of microplastics released from a biodegradable polymer during biodegradation/enzymatic hydrolysis. *Water Res.* 211, 118068.
- Wong, J.K.H., Lee, K.K., Tang, K.H.D., Yap, P.-S., 2020. Microplastics in the freshwater and terrestrial environments: prevalence, fates, impacts and sustainable solutions. *Sci. Total Environ.* 719, 137512.
- Wu, X., Chen, X., Jiang, R., You, J., Ouyang, G., 2022. New insights into the photo-degraded polystyrene microplastic: effect on the release of volatile organic compounds. *J. Hazard. Mater.* 431, 128523.
- Xiao, C., Zhang, M., Ding, L., Qiu, X., Guo, X., 2023. New sight of microplastics aging: reducing agents promote rapid aging of microplastics under anoxic conditions. *J. Hazard. Mater.* 451, 131123.
- Xiao, J., Rabeah, J., Yang, J., Xie, Y., Cao, H., Brückner, A., 2017. Fast electron transfer and $\bullet$ OH formation: key features for high activity in visible-light-driven ozonation with C_3N_4 catalysts. *ACS Catal.* 7 (9), 6198–6206.
- Xiao, R., Noerpel, M., Ling, L., Wei, Z., Spinney, R., 2014. Thermodynamic and kinetic study of ibuprofen with hydroxyl radical: a density functional theory approach. *Int. J. Quant. Chem.* 114 (1), 74–83.
- Xie, Z.-H., He, C.-S., Zhou, H.-Y., Li, L.-L., Liu, Y., Du, Y., Liu, W., Mu, Y., Lai, B., 2022. Effects of molecular structure on organic contaminants' degradation efficiency and dominant ROS in the advanced oxidation process with multiple ROS. *Environ. Sci. Technol.* 56 (12), 8784–8795.
- Xu, Y., Huang, D., Liu, P., Ouyang, Z., Jia, H., Guo, X., 2022. The characteristics of dissolved organic matter release from UV-aged microplastics and its cytotoxicity on human colonic adenocarcinoma cells. *Sci. Total Environ.* 826, 154177.
- Xu, Y., Xian, Z.-N., Yue, W., Yin, C.-F., Zhou, N.-Y., 2023. Degradation of polyvinyl chloride by a bacterial consortium enriched from the gut of *Tenebrio molitor* larvae. *Chemosphere* 318, 137944.
- Yang, Y., Chen, J., Chen, Z., Yu, Z., Xue, J., Luan, T., Chen, S., Zhou, S., 2022. Mechanisms of polystyrene microplastic degradation by the microbially driven Fenton reaction. *Water Res.* 223, 118979.
- Yang, Y., Wang, J., Xia, M., 2020. Biodegradation and mineralization of polystyrene by plastic-eating superworms *Zophobas atratus*. *Sci. Total Environ.* 708, 135233.
- Ye, J., Chen, Y., Gao, C., Wang, C., Hu, A., Dong, G., Chen, Z., Zhou, S., Xiong, Y., 2022. Sustainable conversion of microplastics to methane with ultrahigh selectivity by a biotic–abiotic hybrid photocatalytic system. *Angew. Chem., Int. Ed.* 61 (52), e202213244.
- Yoshida, S., Hiraga, K., Takehana, T., Taniguchi, I., Yamaji, H., Maeda, Y., Toyohara, K., Miyamoto, K., Kimura, Y., Oda, K., 2016. A bacterium that degrades and assimilates poly(ethylene terephthalate). *Science* 351 (6278), 1196–1199.
- Yu, F., Yang, C., Zhu, Z., Bai, X., Ma, J., 2019. Adsorption behavior of organic pollutants and metals on micro/nanoplastics in the aquatic environment. *Sci. Total Environ.* 694, 133643.
- Yuan, J., Ma, J., Sun, Y., Zhou, T., Zhao, Y., Yu, F., 2020. Microbial degradation and other environmental aspects of microplastics/plastics. *Sci. Total Environ.* 715, 136968.
- Zare, Y., Rhee, K.Y., Park, S.J., 2019. Simple model for hydrolytic degradation of poly(lactic acid)/poly(ethylene oxide)/carbon nanotubes nanobiosensor in neutral phosphate-buffered saline solution. *J. Biomed. Mater. Res., Part A* 107 (12), 2706–2717.
- Zhang, A., Hou, Y., Wang, Y., Wang, Q., Shan, X., Liu, J., 2023a. Highly efficient low-temperature biodegradation of polyethylene microplastics by using cold-active laccase cell-surface display system. *Bioresour. Technol.* 382, 129164.
- Zhang, F., Zhao, Y., Wang, D., Yan, M., Zhang, J., Zhang, P., Ding, T., Chen, L., Chen, C., 2021a. Current technologies for plastic waste treatment: a review. *J. Clean. Prod.* 282, 124523.
- Zhang, J., Gao, D., Li, Q., Zhao, Y., Li, L., Lin, H., Bi, Q., Zhao, Y., 2020. Biodegradation of polyethylene microplastic particles by the fungus *Aspergillus flavus* from the guts of wax moth *Galleria mellonella*. *Sci. Total Environ.* 704, 135931.
- Zhang, K., Hamidian, A.H., Tubić, A., Zhang, Y., Fang, J.K.H., Wu, C., Lam, P.K.S., 2021b. Understanding plastic degradation and microplastic formation in the environment: a review. *Environ. Pollut.* 274, 116554.
- Zhang, K., Su, J., Xiong, X., Wu, X., Wu, C., Liu, J., 2016. Microplastic pollution of lakeshore sediments from remote lakes in Tibet plateau, China. *Environ. Pollut.* 219, 450–455.
- Zhang, M., Xiao, C., Ding, L., Wang, T., Guo, X., 2023b. Probing the aging process and mechanism of microplastics under reduction conditions. *J. Hazard. Mater.* 451, 131185.
- Zhang, N., Ding, M., Yuan, Y., 2022a. Current advances in biodegradation of polyolefins. *Microorganisms* 10 (8), 1537.
- Zhang, X., Su, H., Gao, P., Li, B., Feng, L., Liu, Y., Du, Z., Zhang, L., 2022b. Effects and mechanisms of aged polystyrene microplastics on the photodegradation of sulfamethoxazole in water under simulated sunlight. *J. Hazard. Mater.* 433, 128813.
- Zhang, Z., Peng, H., Yang, D., Zhang, G., Zhang, J., Ju, F., 2022c. Polyvinyl chloride degradation by a bacterium isolated from the gut of insect larvae. *Nat. Commun.* 13 (1), 5360.
- Zhao, M., Zhang, T., Yang, X., Liu, X., Zhu, D., Chen, W., 2021. Sulfide induces physical damages and chemical transformation of microplastics via radical oxidation and sulfide addition. *Water Res.* 197, 117100.
- Zhu, B., Ye, Q., Seo, Y., Wei, N., 2022a. Enzymatic degradation of polyethylene terephthalate plastics by bacterial Curli display PETase. *Environ. Sci. Technol. Lett.* 9 (7), 650–657.
- Zhu, K., Jia, H., Sun, Y., Dai, Y., Zhang, C., Guo, X., Wang, T., Zhu, L., 2020a. Long-term phototransformation of microplastics under simulated sunlight irradiation in aquatic environments: roles of reactive oxygen species. *Water Res.* 173, 115564.
- Zhu, K., Sun, Y., Jiang, W., Zhang, C., Dai, Y., Liu, Z., Wang, T., Guo, X., Jia, H., 2022b. Inorganic anions influenced the photoaging kinetics and mechanism of polystyrene microplastic under the simulated sunlight: role of reactive radical species. *Water Res.* 216, 118294.
- Zhu, L., Zhao, S., Bittar, T.B., Stubbins, A., Li, D., 2020b. Photochemical dissolution of buoyant microplastics to dissolved organic carbon: rates and microbial impacts. *J. Hazard. Mater.* 383, 121065.